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# Steam reforming of methane: Current states of catalyst design and process upgrading

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ABSTRACT

Methane ($CH_4$) is the major component of currently abundant natural gas and a prominent green-house gas. Steam reforming of methane (SRM) is an important technology for the conversion of $CH_4$ into $H_2$ and syngas. To improve the catalytic activity and coking resistance of SRM catalysts, great efforts (including the addition of promoters, development of advanced supports, and structural modification, etc.) have been made with considerable progress in the past decade. Meanwhile, a series of novel processes have been explored for more efficient and energy-saving SRM. In this scenario, a comprehensive review on the recent advances in SRM is necessary to provide a constructive insight into the development of SRM technology, however, is still lacking. Herein, the improvements in catalyst construction for conventional SRM and the newly developed SRM processes in the past decade are presented and analyzed. First, the critical issues of SRM catalysts are briefly introduced. Then, the recent research advances of the most popular Ni based catalysts and the catalysts based on the other non-noble metals (Co, Cu, Mo etc.) and the efficient but costly noble metals (Au, Pt, Pd, Rh, Ru etc.) are discussed. Furthermore, the development of the representative modified SRM processes, including thermo-photo hybrid SRM, sorbent enhanced SRM, oxidative SRM, chemical looping SRM, plasma and electrical-field enhanced SRM, is demonstrated, and their advantages and limits are compared. Finally, a critical perspective is provided to enlighten future work on this significant area.

## 1. Introduction

With the growing concern of global warming, it is imperative to minimize the emission of green-house gases in the atmosphere. Carbon dioxide ($CO_2$) and methane ($CH_4$) are two important green-house gases [1,2]. Since the radiative forcing produced per molecule $CH_4$ is greater and the infrared window is less saturated in the wavelengths range of radiation absorbed by $CH_4$, the contribution of per mole $CH_4$ to the greenhouse effect is 25 times that of per mole $CO_2$ [3]. Meanwhile, methane ($CH_4$) is the major constituent of the currently abundant natural gas [4]. While the conversion and utilization of $CO_2$ has received tremendous research attention in recent years [5,6], the valorization of $CH_4$, i.e., converting methane into other value-added chemicals, is the start of C1 chemistry and is gaining increasing interest with the intensive exploration of natural gas resources. Hitherto, methane has been transformed into hydrogen ($H_2$) or syngas (a mixture of CO and $H_2$) via steam reforming of methane (SRM) [7], dry reforming of methane (DRM) [8], and partial oxidation of methane (POM) [9,10]. $H_2$ is an essential green energy for the development of our sustainable society [11–13]. And the syngas can be further used for the industrial synthesis of methanol or other organic compounds via Fischer-Tropsch reactions [14]. Besides, the direct conversion of methane into liquid oxygenates via POM [15,16] and the oxidative coupling of methane [17] have been attempted in recent years.

Among the various methane-conversion strategies, SRM is a predominant industrial process for manufacturing hydrogen and syngas (Fig. 1A) [18,19]. It is an endothermic reaction (Eq. (1.1)) between water steam and methane at a high temperature (typically 1023–1223 K) and a high pressure (typically 14–20 atm) [20]. During the process, the water-gas shift reaction (WGS, Eq. (1.2)) could take place at the meantime, which further enhances the production of $H_2$.

$$CH_4 + H_2O \leftrightarrow CO + 3H_2 \quad \Delta H^0_{298} = +206 \text{ kJ/mol} \tag{1.1}$$

$$H_2O + CO \leftrightarrow CO_2 + H_2 \quad \Delta H^0_{298} = -41 \text{ kJ/mol} \tag{1.2}$$

The catalytic reaction between methane and steam was first reported by Neumann and Jacob in 1924 and attracted tremendous interest then

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**Abbreviations:**

| | |
|---|---|
| SRM | steam reforming of methane |
| DRM | dry reforming of methane |
| POM | partial oxidation of methane |
| WGS | water-gas shift |
| SOFC | solid oxide fuel cell |
| SESRM | sorbent enhanced steam reforming of methane |
| OSRM | oxidative steam reforming of methane |
| CLSRM | chemical looping SRM |
| PSRM | photocatalytic SRM |
| TPSRM | thermo-photo hybrid SRM |
| S/C | steam-to-carbon |
| DFT | density functional theory |
| FSP | flame spray pyrolysis |
| YSZ | yttria-stabilized zirconia |
| DBD | dielectric-barrier discharge |
| DSS | daily start-up and shut-down |
| CH-M | $Ce_{0.65}Hf_{0.25}M_{0.1}O_{2-\delta}$ |
| CZP | $(Ce_{0.75/1.025}Zr_{0.25/1.025}Pr_{0.025/1.025})O_{2-y}$ |
| CZO | $(Ce_{0.75}Zr_{0.25})O_{2-y}$ |
| NCT | $NaCeTi_2O_6$ |
| LSF | $La_{0.6}Sr_{0.4}FeO_{3-\delta}$ |
| STF | $SrTi_{0.7}Fe_{0.3}O_{3-\delta}$ |
| SSZ | scandia-stabilized zirconia |
| SMSI | strong metal-support interaction |
| BET | Brunauer–Emmett–Teller |
| RWGS | reverse water-gas shift |
| FAl | $Al_2O_3$ synthesized by flame spray pyrolysis |
| SAl | Al particle suspension |
| HAP | hydroxyapatite |
| NCASC | Ni supported on calcium aluminate modified SiC |
| LDH | Layered double hydroxides |
| SPMR | steam plasma methane reforming |
| CSCM | combined sorbent-catalyst material |
| ATR | autothermal reforming process |
| FR | fuel reactor |
| SR | steam reactor |
| AR | air reactor |
| SE-CLSRM | sorption enhanced chemical looping methane steam reforming |
| GDC | $Ce_{0.9}Gd_{0.05}Pr_{0.05}O_2$ |
| LSC | $La_{0.8}Sr_{0.2}CrO_3$ |
| RP | Ruddlesden-Popper |
| XRD | in-situ X-Ray diffraction |
| XPS | in-situ X-ray photoelectron spectroscopy |
| XANES | X-ray absorption near edge structure |
| EXAFS | extended X-ray absorption fine structure |

[22]. Soon in 1930, the first industrial application of SRM was implemented [23]. But the research and industrial efforts in improving the catalyst performance and the reactor tube materials have never stopped. Till now, SRM remains an important topic in the scientific field (Fig. 1B). Developing more efficient and more cost-effective SRM technology is a long-term project [24,25]. Among the various explored catalysts for SRM, $Ni/Al_2O_3$ is the most widely employed due to its low cost and high activity [26,27]. However, coking formation and sintering of Ni particles are two main challenges for the commercial Ni catalyst. In order to obtain catalysts with high coking and sintering resistances, considerable research efforts have been paid on designing advanced Ni catalysts, such as promoter-modified Ni catalysts, solid-solution catalysts, novel supported Ni catalysts, and self-supported Ni catalysts [28–31]. Besides, exploring the potential of non-Ni-based catalysts is also important to provide alternatives for future industry. Thereby, a number of researchers have been trying to improve the performance and practicability of other non-noble metal (Co, Cu, Mo) catalysts [32–34] and noble metal (Ru, Rh, Pt) catalysts [35–37].

Moreover, to improve the SRM efficiency with reduced overall energy input, a series of advanced technologies have been developed, including sorbent enhanced SRM (SESRM) [38], oxidative SRM (OSRM) [20,39,40], chemical looping process [41], photocatalytic SRM [42], thermo-photo hybrid SRM [43], solid oxide fuel cell (SOFC) [44], plasma SRM [45], and electro-catalytic SRM [46] have been explored, respectively. But a systematic analysis on the advantages and disadvantages of all the methods is lacking.

Given the significance of SRM for the effective utilization of $CH_4$ and production of $H_2$ and syngas, as well as the continuous progresses of SRM in the past decade, an up-to-date overview of the pertinent findings for SRM is necessary to provide sufficient insights that could lead to great breakthroughs in this area. However, recent reviews on SRM are few and are all small ones focusing on limited aspects. Specifically, Iulianelli et al. reviewed the SRM processes on membrane reactors in 2016 [47]. In 2018, Kaiwen et al. provided an economic analysis of the SRM process [48]. In 2019, Paulina Suma et al. overviewed the recently developed catalytic materials for SRM [49]. Recently, Meloni et al.

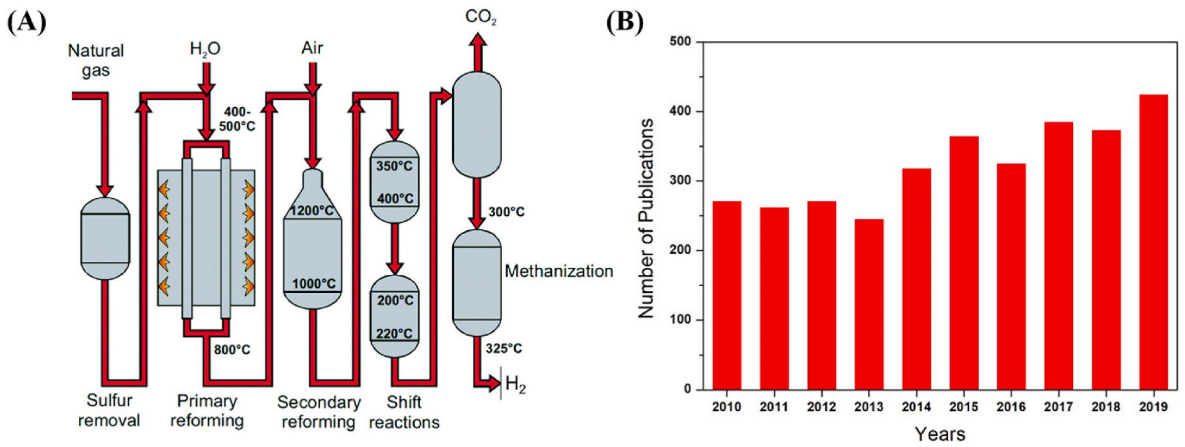



**Fig. 1.** (A) Process flow diagram of steam methane reforming for $H_2$ production. Reproduced with permission from Ref. [21]. Copyright 2020, Royal Society of Chemistry. (B) Publications on SRM in the past decade.

reported a short review on Ni-based SRM catalysts [50]. Thereby, in this work, a comprehensive picture about the recent development of SRM catalysts and technologies is exhibited. First, the recent advances of catalysts based on Ni, other non-noble metals (Co, Cu and Mo) and noble metals (Ru, Rh, Pt, etc.) for SRM are outlined. Furthermore, advanced SRM processes, which can increase the $H_2$ selectivity, reduce the energy input or achieve better product separation, are discussed with an emphasis on their advantages and disadvantages. Finally, the future research directions of catalysts for SRM and perspective for the development of advanced SRM processes are provided.

## 2. Key issues of SRM catalysts

An ideal SRM catalyst should possess both a high activity and a long-term stability. These performances of a catalyst are highly related to its chemical nature and textural properties (morphology, surface area, pore structure etc.). However, coking and sintering have been the two major issues that lead to the deactivation of a catalyst during the SRM process [27,51,52].

Sintering is the growing of metal particles (Fig. 2A). It is a complex process influenced by many factors such as temperature, chemical environment, catalyst composition, structure, and support morphology, among which temperature and atmosphere are the two most important factors [27]. For instance, a higher temperature results in an obviously higher sintering rate. Water steam atmosphere can accelerate the sintering as well. Two mechanisms are proposed to describe the sintering: (1) atom migration (Ostwald ripening) and (2) crystallite migration and coalescence. Ostwald ripening mechanism involves the metal atoms emitting from one particle to another, while the crystallite migration and coalescence process indicates that crystallites move over the support and collide each other to form larger particles. The driving force for sintering is the difference in the surface energies before and after sintering. Sintering leads to the reduction of metal surface area, thus resulting in the decreased activity [18]. Sintering can be effectively suppressed by enhancing the metal-support interaction. For example, the addition of Ir or Rh can mitigate the sintering of Ni by forming Ni–Ir and Ni–Rh alloys during the aging process, respectively [53]. The application of $MgAl_2O_4$ as the support of Rh catalyst led to a strong metal-support interaction with the formation of Rh–O bonds [54]. The modification of structural properties, such as the mesoporous structure, can also help mitigate the sintering [55].

Coking is a collective description of various kinds of carbonaceous deposit formed in the reactor which is originated from the dissociation of hydrocarbons on the surface of the catalysts (Fig. 2B). These carbon species, which are probably atomic carbon ($C_\alpha$), are highly reactive. Furthermore, some $C_\alpha$ species are converted into $C_\beta$ by rearrangement and polymerization. $C_\beta$ may be gasified or dissolved in the Ni crystallite. The dissolved $C_\beta$ can diffuse through the nickel to nucleate and precipitate at the rear of crystallite, forming carbon whisker, which can lift nickel crystallite from the support and finally result in fragmentation [56,57]. The third type of coking is encapsulating carbon (gum) which is generated during the reforming of heavy hydrocarbon feeds with a high content of aromatic compounds. This kind of carbon consists of a thin $CH_x$ film or a few layers of graphite covering the Ni particle, leading to the loss of activity. The strategies to reduce coking are various. The employment of promoter like alkali metal [58], noble metal [59], and rare earth metal [60] can reduce coke formation. Application of well-defined supports, such as perovskites [61] and spinels [55], can reduce the carbon formation as well.

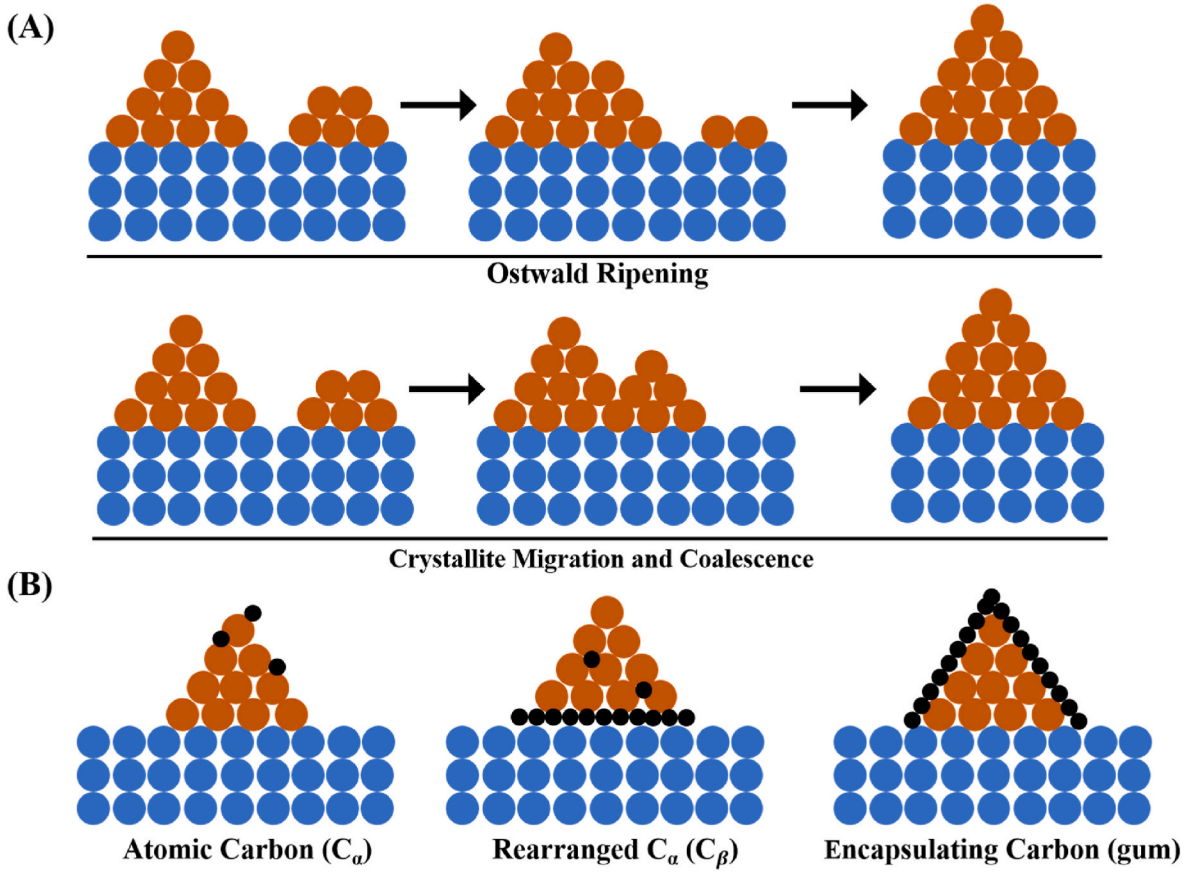



**Fig. 2.** Schematic illustrations of the (A) sintering and (B) coking issues of an SRM catalyst.

## 3. Catalyst progress in conventional SRM

### 3.1. Ni based catalysts

Ni based catalysts have been the most commonly used catalysts for the traditional SRM process due to their low cost and high activity. Since coking and sintering remain the major issues for Ni based SRM catalysts, recent reports on SRM have been focusing on improving the coke and sintering resistance of Ni based catalysts (Table 1). Strategies that have been applied include developing new promoters, exploring Ni-based solid solution catalysts, tuning the structures of supported Ni catalysts and constructing self-supported Ni catalysts.

#### 3.1.1. Promoters for Ni based catalysts

Various species have been explored as promoters for Ni based catalysts, including noble metals, coinage metals, redox metal oxides, non-metals, and so forth.

**1) Promoters Based on Metal-metal Interaction**

A series of metals can function as promoters for Ni catalysts in SRM, including noble metals (like Pd, Rh, Ir, Ru, and Pt), coinage metals (i.e. Au, Ag, and Cu), rear earth metals (represented by Ce), alkali metal (e.g. K) etc.

Most noble metals could promote the performance of Ni based catalysts in SRM. With the addition of Pd, yttria-stabilized zirconia (YSZ) supported Ni catalysts reached a high $CH_4$ conversion of 94.6% at 650 °C, though the CO selectivity is relatively low due to the promoted WGS by Pd [83]. Rh and Ir could enhance both the activity and sintering resistance of $Ni/Al_2O_3$ by forming Rh–Ni and Ir–Ni alloy, respectively [53]. However, the formation of Ru–Ni was energetically unfavorable due to the lower miscibility of Ru in Ni and poorer sintering resistance under an aging condition of 800 °C [53]. As demonstrated in Fig. 3, even though the initial activity over $Ni–Ru/Al_2O_3$ was comparable to that over $Rh–Ni/Al_2O_3$ and superior to that over $Ni–Ir/Al_2O_3$ (Fig. 3A), $Ni–Ru/Al_2O_3$ showed the worst performance among the three after aging at 800 °C (Fig. 3B) [53].

Nevertheless, Ru as well as Pt, in a small loading (<0.1 wt%), could promote the self-reducibility of $Ni/MgAl_2O_4$ catalyst during the SRM process via hydrogen spillover on Ru [123–125]. Doping of 0.05 wt% Ru or Pt could endow an anodic-alumina-supported Ni (17.9 wt%) catalyst, which deactivated quickly in daily start-up and shut-down (DSS) SRM at 700 °C, with the ability of self-activation, self-regeneration, and self-redispersion [126]. Moreover, recently core-shell structure Ni@Pt was reported to significantly suppress the carbon formation and double the activity [127]. The modified electron structure of Ni by Pt provided a down-shifted d-band which could accelerate the C–O bond association between *OH and $*CH_x$ species, thus enhancing the coking resistance.

Among the coinage metals (i.e. Au, Ag, Cu), Cu has been recognized as an excellent promotor for enhancing the CO selectivity and stability of $Ni/Al_2O_3$ [128].Recent attention on Ag and Au revealed that while the two metal species could enhance the stability of Ni based catalysts, decreased $CH_4$ conversion could be resulted at the meantime [129,62]. This is because Ag and Au preferentially alloyed with low-coordinated active Ni sites, increased the activation energy for $CH_4$, and thus reduced the overall performance of the catalyst [62,130]. Besides, both the Ni–Au and Ni–Ag alloys showed higher energy barrier than Ni for the oxidation of *CH to *CHO [131]. Similarly, while the $CH_4$ conversion kept at ~95% for 180 h on $Ni–Co/Al_2O_3$ bimetallic catalyst with 7% Co, the low-coordinated active Ni sites could be partially blocked by Co atoms, leading to decreased activity with increasing Co content [64].

Ce could promote the activity of $Ni/ZrO_2$ catalyst because $CeO_2–ZrO_2$ had an excellent oxygen storage/release capacity, which could promote the coking resistance of the catalyst [132]. Also, the interaction between Ni and $Ce–ZrO_2$ support could improve the reducibility of Ni and Ce [63]. The catalytic performance highly depends on the Ce:Zr ratio [82]. The $CH_4$ conversion reached 97% when the ratio of Ce:Zr was 1, and decreased to ~71% (which was even lower than pure $ZrO_2$ support) when the ratio of Ce:Zr was 5.

K enhanced the coking resistance of Ni via increasing the C–H cleavage barriers in the last two steps of $CH_4$ dissociation [133]. But the activity of the K promoted Ni catalyst was only 17% of the undoped $Ni/Al_2O_3$ [58]. In contrast, the co-addition of K and Ti species could significantly improve the SRM activity of $Ni/Al_2O_3$ catalyst, as demonstrated in Fig. 3C [65].

**2) Redox Metal Oxide Promoters**

In the past decade, besides metals, redox metal oxides, such as $CeO_2$, $La_2O_3$, $Nb_2O_5$ etc., have also been found effective in stabilizing the Ni particles against sintering at high temperatures [66,134,135]. With the presence of $CeO_2$ on $Ni–CeO_2–Al_2O_3$/cordierite catalysts, not only the methane conversion and the stability were improved, but also the $CO_2$ content in the outlet gas was considerably decreased (which might be attributed to suppression of the WGS reaction) [136]. The optimal amount of $CeO_2$ promoter could vary with the Ni loading amounts in $Ni/Al_2O_3$. For 13 wt% $Ni/Al_2O_3$, 1.02 wt% Ce showed the best promotion effect, maintaining 75% $CH_4$ conversion at an S/C ratio of 2.7 in 300 h [52].

Nb improved the SRM performance of $Ni/Al_2O_3$ catalyst by suppressing the coke formation on the catalyst via their strong interaction with both the metal species (Ni) and the support ($Al_2O_3$) (Fig. 3D) [66]. With 5 wt% Nb loading, the $CH_4$ conversion on $Ni/Al_2O_3$ reached 98%, which was 10.4% higher than that of the pristine sample. However, when the Nb content reached 20 wt%, the aggregation of Ni became severe and the catalytic activity was reduced.

When $CaZrO_3$ perovskite oxide was applied as a promoter to enhance the catalytic performance of $Ni/\alpha-Al_2O_3$, the size and amount of $CaZrO_3$ affected not only the dispersion of Ni, but also the interaction between Ni and $Al_2O_3$ and the number of oxygen vacancies [67]. Promoted by 15 wt% $CaZrO_3$, $Ni/\alpha-Al_2O_3$ attained a $CH_4$ conversion of 67% at 700 °C with an S/C ratio of 1.0.

**3) Non-metal Promotors**

Boron is the only non-metal promoter that has been explored so far [137]. It could enhance both the activity and the stability of Ni based catalysts via preventing the coke formation. While the pristine catalyst could only convert 55% $CH_4$ at the beginning and lost 21% of its initial activity after 10 h SRM reaction, the $CH_4$ conversion of 1 wt% B doped catalyst reached ~61% at the beginning and kept at ~56% after 10 h. However, the excessive addition of B could reduce the activity of Ni catalysts, because the B atoms could block all the step sites of Ni first and then occupy the octahedral sites just below the surface.

#### 3.1.2. Ni based solid solution catalysts

Ni based solid solution catalysts, represented by Ni/MgO, are a special type of catalysts with the active Ni species reduced from the parent solid solution, which allowed the formation of fine Ni particles with extraordinarily high stability [138]. $Ni_{0.03}Mg_{0.97}O$ solid solution catalyst with small and well-dispersed Ni metallic particles on the surface attained 90% $CH_4$ conversion at a low S/C ratio of 1.0 and kept stable for 70 h with a high coke resistance [139]. The metallic Ni particles which homogeneously distributed on the surface of the $Ni_{0.4}Mg_{0.6}O$ solid solution supports (Fig. 4A) completely converted $CH_4$ for 1000 h without deactivation [68]. Furthermore, tri-compound Ni–Mg–Al catalysts have been engineered to drive efficient SRM at a wide range of S/C ratio from 1.0 to 4.0. Even at an S/C ratio as high as 4.0 and a temperature as low as 550 °C, 100% conversion of $CH_4$ could still be realized on Ni–Mg–Al catalysts [70]. Also, $Ni_{0.5}Mg_{2.5}AlO_9$ catalysts showed higher activity and stability than those of $Ni/ZrO_2/Al_2O_3$ and $Ni/La–Ca/Al_2O_3$ with an extremely short residence time of 20 ms [140].

#### 3.1.3. Structure regulation of supported Ni catalysts

**1) Ni Particle Size Control**

The coking resistance and catalytic activity were both intimately affected by the Ni particle size. A recent study with Ni particles on $SiO_2$

**Table 1**
Catalysts investigated for SRM in the past decade.

| Catalyst | Metal Loading (wt%) | Promoter | Temperature (K) | $CH_4$ conversion (%) | S/C ratio | Reference |
|---|---|---|---|---|---|---|
| Ni/α-$Al_2O_3$ | 2.9 | Rh | 773 | | 4 | [53] |
| | | Ir | | | | |
| | | Ru | | | | |
| Ni/$Al_2O_3$ | 15 | Au | | | | [62] |
| Ni/$Al_2O_3$ | 13 | Ce | 1073 | 75 | 2.7 | [52] |
| Ni//$Al_2O_3$ | 5 | $CeO_2$–$ZrO_2$ | 923 | ~63 | 2 | [63] |
| Ni/γ-$Al_2O_3$ | 12 | Co | 1023 | 95 | 2 | [64] |
| Ni/α-$Al_2O_3$ | 12.7 | K | | | | [58] |
| Ni/$Al_2O_3$ | 10 | $K_2Ti_xO_y$ | 1023 | 97.2 | 2.5 | [65] |
| Ni/$Al_2O_3$ | 15 | Nb | 1023 | 98 | 1 | [66] |
| Ni/α-$Al_2O_3$ | 15 | $CaZrO_3$ | 923 | 67 | 1 | [67] |
| $Ni_{0.4}Mg_{0.6}O$ | | | 1023 | ~100 | 0.5 | [68] |
| Ni/MgO | | | 1023 | ~90 | 3 | [69] |
| Ni/MgO–$Al_2O_3$ | 12.5 | | 923 | | 3 | [70] |
| Ni–MgAl(CO-IM) | 13.4 | | 1073 | ~75 | 1.24 | [71] |
| Ni/$MgAl_2O_4$ | 15.3 | | 823 | 50 | 5 | [72] |
| Ni/$MgAl_2O_4$ | 15 | Pt | 823 | ~65 | 5 | [73] |
| Ni/$SiO_2$ | 10 | | 923 | ~46 | 1.5 | [74] |
| Ni/$SiO_2$ | 10 | | 1023 | ~40 | 0.5 | [75] |
| Ni/Si-AE | 5 | | 973 | ~90 | 1 | [76] |
| Ni/$SiO_2$/$Al_2O_3$ | 10 | | 923 | 96 | 3.5 | [77] |
| NiWSi/FT | 5 | Ce | 923 | 73 | 3 | [78] |
| NCASC | 10 | CaO/$Al_2O_3$ | 1073 | 97.7 | 3 | [79] |
| Ni@yolk-$ZrO_2$ | 5 | | 973 | 93 | 2.5 | [80] |
| Ni@yolk-$ZrO_2$ | 5 | | 973 | 90 | 2.5 | [81] |
| Ni/$Ce_{0.85}Zr_{0.15}O_{2-\delta}$ | 5 | | 823 | ~71 | 1 | [82] |
| Ni/YSZ | | Pd | 873 | 94.6 | 3 | [83] |
| Ni/$La_2Sn_2O_7$ | 12 | | 1023 | 30 | 2 | [84] |
| Ni/$La_2Zr_2O_7$ | | | | 98 | | |
| Ni/$Y_2Zr_2O_7$ | 10 | | 1023 | 94 | 2 | [85] |
| Ni/$Y_2Ti_2O_7$ | 7 | | 1023 | ~96 | 2 | [86] |
| Ni/$Y_2Sn_2O_7$ | | | | ~88 | | |
| Ni/$Y_2Ce_2O_7$ | | | | ~88 | | |
| Ni/$La_2Ti_2O_7$ | 7 | | 1023 | ~88 | 2 | [87] |
| Ni/$Pr_2Ti_2O_7$ | | | | ~90 | | |
| Ni/$Sm_2Ti_2O_7$ | | | | ~92 | | |
| Ni/$NaCeTi_2O_6$ | 4 | | 973 | 91 | 1 | [88] |
| Ni-LSF/STF | 10 mol% | | 823 | 90 | 1 | [89] |
| Ni/$LaFeO_3$ | 12 | | 1073 | ~80 | 2 | [90] |
| Ni/$LaTiO_5$ | 15.55 | | 1173 | 95 | 1 | [24] |
| $NiLaO_3$ | 6.2 | Au | 923 | 40 | 1.24 | [91] |
| Ni/FeCr | | | 923 | 97.4 | 3 | [92] |
| Ni/CH | | | 923 | 85 | 2 | [93] |
| Ni/ChH-Pr | | | | 86 | | |
| Ni/γ-$Al_2O_3$ | 10 | | 1023 | 97 | 2 | [94] |
| Ni/MgO–$Al_2O_3$ | | | | 97.2 | | |
| Ni/$CeO_2$ | | | | 89 | | |
| Ni/$Ce_{0.4}Zr_{0.6}O_2$ | | | | 91 | | |
| Ni/$Al_2O_3$ honeycomb | 20.2 | Ce | 873 | 79.1 | 2 | [95] |
| Ni honeycomb | | | 1073 | 97 | 1.34 | [96] |
| Ni honeycomb | | | 1023 | 50 | 1.36 | [97] |
| $Ni_3Al$ | | | 973 | 65 | 1 | [98] |
| Ru/La–$Al_2O_3$ | 1.5 | | 1023 | 97.3 | 3 | [99] |
| Ru/$Al_2O_3$@Al | 2 | | 973 | ~97 | 3 | [100] |
| Ru/$Ni_6Al_2$ | 0.5 | | 1023 | ~98 | 1 | [101] |
| Ru/$Co_6Al_2$ | 1 | | 1023 | 100 | 3 | [102] |
| Ru/MgO | 2.4 | | 973 | 99 | 4 | [103] |
| Ru/$Nb_2O_5$ | 3 | | 973 | 100 | 4 | |
| Ru/$CeZr_{0.5}GdO_4$ | 3 | | 1023 | 97 | 3 | [104] |
| Rh/$Al_2O_3$ | 1 | | 823 | 69 | 4 | [105] |
| Rh/$MgAl_2O_4$ | 5 | | 1123 | ~41 | 3 | [54] |
| Ir/$MgAl_2O_4$ | | | | ~55 | | |
| Rh/HAP | 1 | | 973 | 77 | 3 | [106] |
| Pd/$CeO_2$ | 3 | | | 75 | | [107] |
| Pt/$CeO_2$–$La_2O_3$–$Al_2O_3$ | 1 | | 1073 | | 3 | [108] |
| Pt/$CeO_2$ | 10 | | | 71.4 | | [109] |
| | | Pd | | | | |
| | | Ir | | | | |
| | | Ru | | 30 | | |
| Co–Pt–Zr–La/Al2$O_3$ | 5 | | 973 | 99.3 | 1.25 | [110] |
| Cu/$Co_6Al_2$ | 5 | | 973 | 96 | 3 | [33] |
| $CeO_2$ | | | 973 | | 0.1 | [111] |
| $Ce_{0.9}Gd_{0.1}O_2$ | | | | | | |
| $Ce_{0.9}Pr_{0.1}O_2$ | | | | | | |

(*continued on next page*)

**Table 1** (*continued*)

| Catalyst | Metal Loading (wt%) | Promoter | Temperature (K) | $CH_4$ conversion (%) | S/C ratio | Reference |
|---|---|---|---|---|---|---|
| $Ir/Ce_{0.9}Gd_{0.05}Pr_{0.05}O_2$ | 0.1 | | 923 | 74 ($H_2O$ conversion) | 0.22 | [112] |
| $La_{0.8}Sr_{0.2}Fe_{0.8}Cr_{0.2}O_3$ | | | 923 | 97 | 3 | [113] |
| $NiAl_2O_4/Al_2O_3$ | 17 | | 823 | ~50 | 3 | [114] |
| $Rh/CeO_2$ | 1.5 | | 1023 | ~100 | 1.2 | [115] |
| $Pt/CeO_2$ | 1.5 | | | | | |
| $Ni/CeO_2$ | 7.5 | | | | | |
| $Ni/Al_2O_3$ | 7 | | 773 | 75 | 4 | [116] |
| $Ni/Al_2O_3$ | | Au | | 84 | | |
| $Ni/Al_2O_3$ | | Ag | | 2.5 | | |
| $Ni/La_2O_3–Al_2O_3$ | | | | 74.2 | | |
| $Ni/CeO_2–Al_2O_3$ | | | | 82 | | |
| $NiAl_2O_4/Al_2O_3$ | 17 | | 873 | 80 | 3 | [117] |
| $Ni/MgAl_2O_4$ | 20 | La | 823 | | 3 | [118] |
| | | Rh | | | | |
| | | B | | | | |
| $Ni/Ce_{0.8}Zr_{0.2}O_2$ | 15 | | 823 | 59.5 | 1 | [119] |
| $Ni/TiO_2$ | 10 | | 723 | 45 | 3 | [120] |
| | 20 | | | 31 | 1 | |
| $Ni/Ce_x$-$Ti_yO_2$ | 10 | | 723 | 63 | 1 | [121] |
| $Ni/La_2O_3–ZrO_2$ | 10 | | 723 | 41.2 | 3 | [122] |
| $Ni/La_2O_3–CeO_2–ZrO_2$ | 10 | | | 32.9 | | |
| $Rh/La_2O_3–ZrO_2$ | 1 | | | 24.3 | | |
| $Rh/La_2O_3–CeO_2–ZrO_2$ | 1 | | | 20.8 | | |

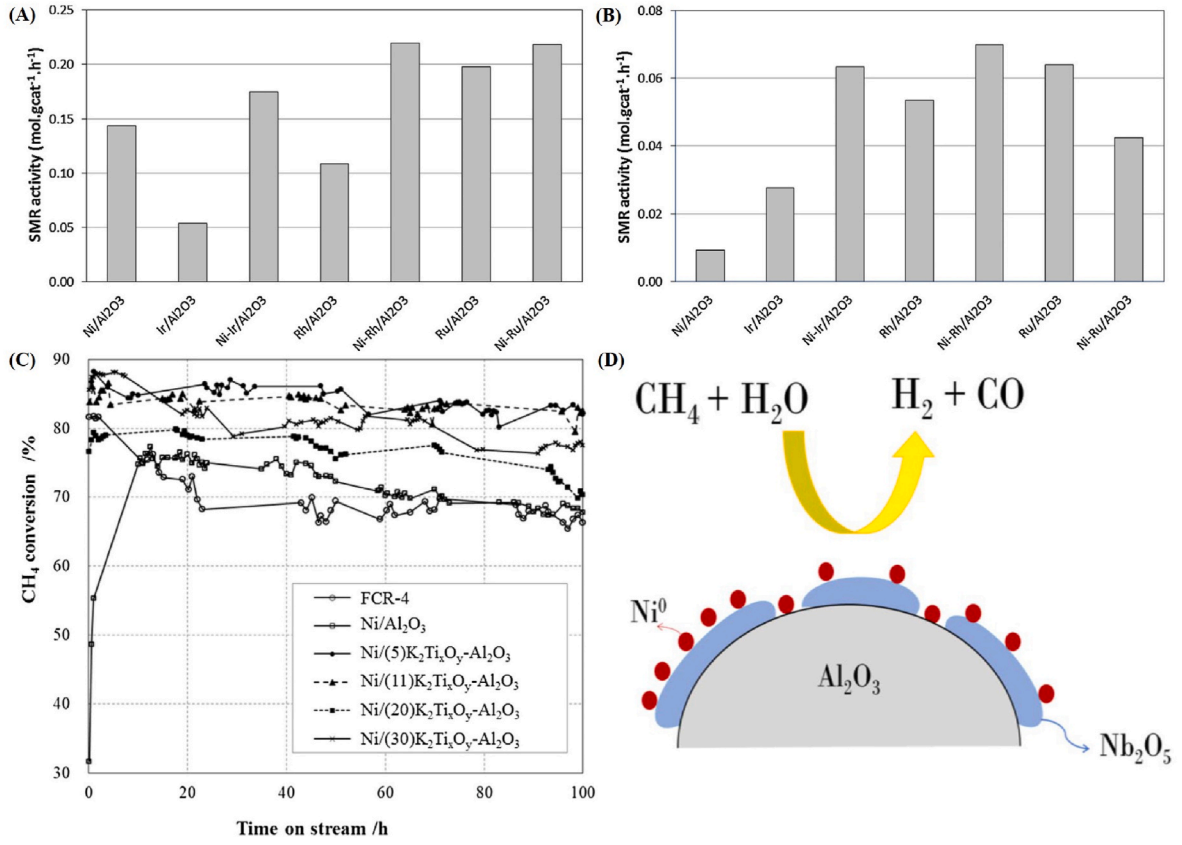



**Fig. 3.** Catalytic activity measured of (A) reduced and (B) aged noble-metal-promoted catalysts. Reproduced with permission from Ref. [53]. Copyright 2015, Elsevier. (C) Change in activity of $Ni/K_2Ti_xO_y$-$Al_2O_3$ catalysts and reference catalysts with time-on-stream. Reproduced with permission from Ref. [65]. Copyright 2014, Elsevier. (D) Scheme structure of Nb-promoted $Ni/Al_2O_3$. Reproduced with permission from Ref. [66]. Copyright 2018, Springer Nature.

in a size range from 1.2 to 6.0 nm suggested that carbon whisker formation was most serious with Ni at 4.5 nm and the optimal activity was achieved on 2–3 nm Ni [141]. Besides, strong interaction could form between small-size (5–10 nm) NiO particles and $CeZrO_2$ support, leading to fast and easy oxygen transfer to and from $NiO/Ni^0$ active phases. Thus, 250 h stable SRM could be realized at an S/C ratio of 2.0 on the catalyst [142].

The crystallite size of Ni or NiO on a supported catalyst is dependent on the synthesis methods, conditions (e.g. calcination temperature), and the loading amount of Ni. Polymer-assisted method allowed the formation of small Ni particles (~8.7 nm) on Si-AE support due to the chelation effect introduced by polymer during calcination, while 45.5 nm Ni particles were formed on the same support via incipient wetness impregnation method [76]. As a result, even though with a weaker

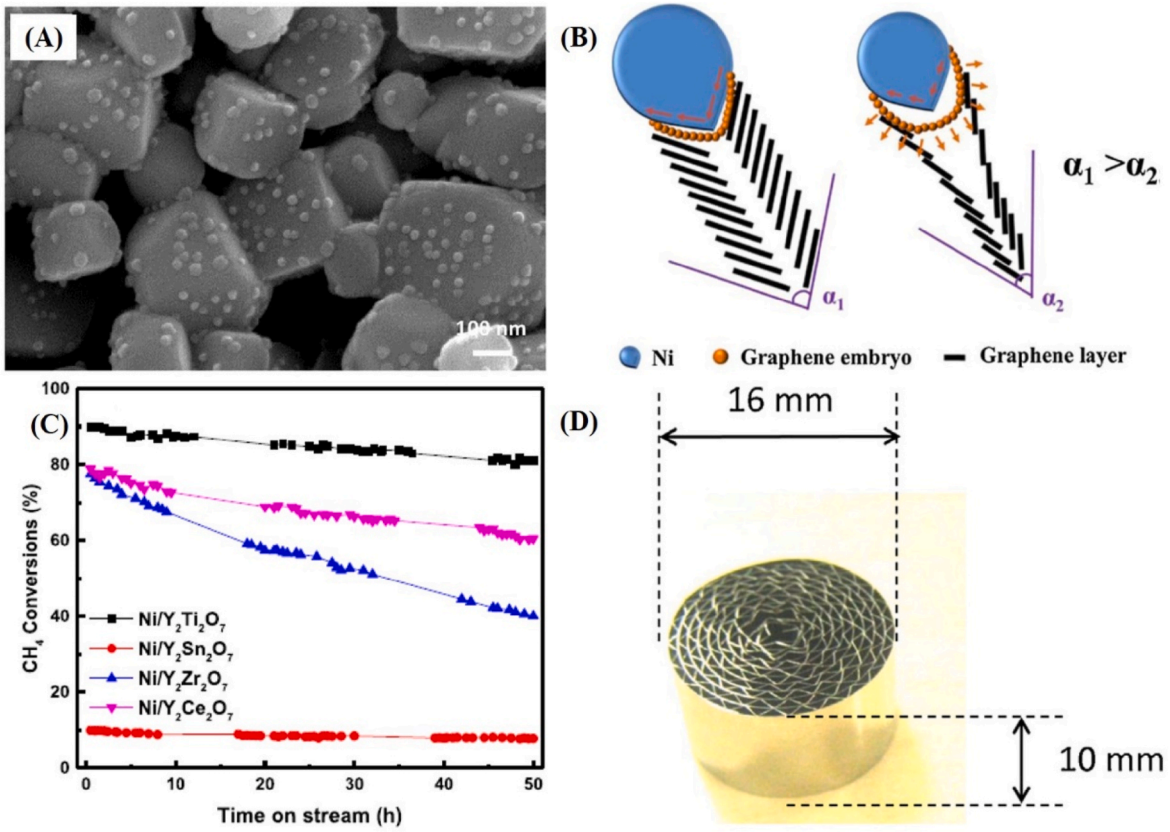



**Fig. 4.** (A) SEM images of reduced NiO–MgO catalyst samples. Reproduced with permission from Ref. [68]. Copyright 2018, Elsevier. (B) Schematic representation of carbon deposition during SRM reaction over Ni/$SiO_2$–C (left) and Ni/$SiO_2$-DBD (right) catalysts. Reproduced with permission from Ref. [75]. Copyright 2015, Elsevier. (C) Long-term stability tests of Ni/$Y_2B_2O_7$ (B = Ti, Sn, Zr, or Ce) catalysts at 750 °C for methane steam reforming. Reproduced with permission from Ref. [86]. Copyright 2018, Elsevier. (D) The structure of Ni honeycomb. Reproduced with permission from Ref. [96]. Copyright 2015, Elsevier.

NiO-support interaction, the sample prepared by polymer-assisted method showed a higher and more stable $CH_4$ conversion and $H_2$ yield with little carbon deposition in 40 h. With 5–15% loading, the NiO on NiO–$SiO_2$ catalyst prepared by a sol-gel method was 9–15 nm in size after calcination at 350–500 °C [77]. In addition, by dielectric-barrier discharge (DBD) treatment, a high Ni dispersion and a small Ni particle size could be attained [75,74]. The smaller Ni particles formed a smaller angle with graphene embryo and significantly prohibited coking (Fig. 4B).

**2) Support Morphology Tuning**

The morphologies of the supporting species as well as their relative location with the active components could shed significant influence on the catalytic performance in SRM. Core/shell structure of Ni@$SiO_2$ with Ni nanoparticle as the core and $SiO_2$ layer as shell has received increasing attention in these years. The $SiO_2$ shell was believed to provide physical protection to Ni nanoparticles, avoiding the aggregation of the Ni particles. Compared with other silica-supported, Ni@$SiO_2$ exhibited a higher $CH_4$ conversion rate with a low Ni loading and a low S/C ratio. A Ni@$SiO_2$ catalyst synthesized by a deposition-precipitation method exhibited a high methane conversion of 85% at 750 °C [143]. Similarly, a Ni@$SiO_2$ catalyst consisted of a 10–15 nm Ni nanoparticle as the core and 30 nm-thick $SiO_2$ shell achieved 83% $CH_4$ conversion at 800 °C at an S/C ratio of 1.0 [144].

On the other hand, Zhang et al. suggested that the introduction of a silica layer in-between $Al_2O_3$ and Ni could avoid the formation of the inactive $NiAl_2O_4$ spinel phase and thus improve the catalytic performance. Compared to Ni/$Al_2O_3$ in Sil-1 shell, the Ni/$Al_2O_3$-Sil-1 core–shell catalyst with Ni deposited on a preformed $Al_2O_3$-Sil-1 core-shell beads exhibited 10% enhancement in SRM activity along with a high stability [145].

Another interesting morphology of the support is hollow shell, which is not only high in surface area but could also provide protective confinement effect. Typically, Lim et al. synthesized a unique hollow shell structure of $ZrO_2$ [80]. Metallic Ni particle with particle size of 8.9 nm could be uniformly dispersed on the $ZrO_2$ hollow shells with suppressed aggregation. Besides, the large amount of micropores on the shell allowed efficient gaseous exchange. As a result, 93% $CH_4$ could be converted over this specially structured catalyst at 700 °C at an S/C ratio of 2.5 with only a slight degree of deactivation in 20 h. Similarly, the rich porosity and appropriate pore sizes of Ni/yolk-$ZrO_2$ catalyst contributed greatly to its high activity [81]. Besides, synthesized via sucrose-concentrated $H_2SO_4$ dehydration reaction, flake-shaped NiO-YSZ outperformed the mixed commercial NiO-YSZ in SRM [146]. However, it was found that the strong interaction between NiO and the substrate hampered the reduction of NiO.

Furthermore, an additional protection shell for the Ni-based catalysts could further increase its activity. For instance, zeolites, which have been widely explored as supports for various catalysts [147,148], can also serve as effective protection shells for Ni-based catalysts. H-β zeolite membrane encapsulated 1.6%Ni/1.2%Mg/$Ce_{0.6}Zr_{0.4}O_2$ prepared by a physical coating method showed 2-3-factor enhancement in $CH_4$ conversion compared to the catalyst without a zeolite shell [149]. The promotion was contributed by both the confined reaction effects (increase residence time within pores) and the promotion effect of $Al^{3+}$ in the zeolite shell to the active sites.

### *3.1.4. Emerging metal oxide substrates*

Supporting materials have been extensively explored all through the development of SRM technology. In the past decade, due to their rich chemical properties, complex oxides with fixed stoichiometric ratios,

such as pyrochlore ($A_2B_2O_7$), perovskite ($ABO_3$), and other complex metal oxides have attracted increasing attention as potential substrates for SRM catalysts [86,84,85,87].

$La_2Sn_2O_7$ and $La_2Zr_2O_7$, as supports, could significantly improve the stability of Ni based catalysts despite of a low $CH_4$ conversion and $H_2$ yield at the beginning [86,84,85,87]. While Ni/γ-$Al_2O_3$ lost ~15% of its $CH_4$ conversion after 80 h, $La_2Zr_2O_7$ supported catalyst showed increasing $CH_4$ conversion during a long-term running (250 h) and finally reached $CH_4$ conversion of ~95%. $La_2O_2CO_3$ in Ni/$La_2Zr_2O_7$ was found responsible for removing carbon deposition. Ni/$Y_2Zr_2O_7$ showed even better performance with the complete conversion of $CH_4$ at 800 °C for more than 200 h without any carbon deposition. The high activity was mainly attributed to the smaller Ni particle size than Ni/$La_2Zr_2O_7$ and the stronger metal-support interaction (SMSI) between Ni and Y.

Different metal elements (Ti, Sn, and Ce) were investigated as alternatives for Zr in Ni/$Y_2Zr_2O_7$ [86]. Among them, $Y_2Sn_2O_7$ and $Y_2Ce_2O_7$ exhibited the lowest activity for the severe sintering of Ni particles. Since $Y_2Ti_2O_7$ possessed the best dispersion of Ni species and the largest density of oxygen vacancies, it exhibited the best $CH_4$ conversion activity and stability (Fig. 4C).

Furthermore, $A_2Ti_2O_7$ materials with A = La, Pr, Sm, or Y have been investigated as potential substrates for Ni in SRM. Among them, the Y contained substrate allowed the formation of Ni particles with the smallest size and the largest surface area [87]. Thereby, the highest $CH_4$ conversion and $H_2$ yield with lowest $H_2$/CO ratio was obtained on Ni/$Y_2Ti_2O_7$.

Perovskite materials, which show high coking resistance provided by their rich surface oxygens, have a relatively long history as supports for Ni catalyst in studies [61]. Ni/$NaCeTi_2O_6$ (Ni/NCT) showed a high activity with a $CH_4$ conversion of ~95% at 700 °C without deactivation for 24 h [88]. Ni/$LaFeO_3$ showed $CH_4$ conversion of ~80% and $H_2$ yield of ~70% at 800 °C at an S/C ratio of 2.0 [90]. Two complex Ni/perovskite catalysts were evaluated by Penner and co-workers [89]. By forming hollow shells on $La_{0.6}Sr_{0.4}FeO_{3-\delta}$ (LSF) and $SrTi_{0.7}Fe_{0.3}O_{3-\delta}$ (STF) substrates, the Ni particles were free from sintering during the reduction process. Strong interaction between Ni and La on $NiLaO_3$ perovskite could prevent the sintering of Ni as well. With a strong metal-support interaction (SMSI), Ni/$La_2TiO_5$, which was derived from ordered double perovskite $La_2NiTiO_6$, reached a $CH_4$ conversion of 95% at 950 °C and maintained it for 24 h [24].

Other complex oxide substrates, such as complex $Ce_{0.65}Hf_{0.25}M_{0.1}O_{2-\delta}$ (CH-M, M = Tb, Sm, Nd, Pr, and La) solid solutions have also been fabricated and used as supports for Ni catalyst [93]. The introduction of Pr, Tb, and La increased the amount of oxygen vacancies and facilitated their coke resistance. Ni/CH-Pr was the best among the studied catalysts with a $CH_4$ conversion of 85% and a long-term stability for 30 h, probably due to its highest oxygen storage capacity and enhanced oxygen mobility. Analogously, Ni catalysts on $(Ce_{0.75/1.025}Zr_{0.25/1.025}Pr_{0.025/1.025})O_{2-y}$ (Ni-CZP), which was a mixture of a Pr-rich λ phase and a Ce-rich cubic phase, demonstrated much better catalytic performance than that of Ni catalysts supported on $(Ce_{0.75}Zr_{0.25})O_{2-y}$ (Ni-CZO), due to the strained but coherent oxygen-vacancy-rich interface between the λ phase and the cubic phase [150].

### 3.1.5. Self-supported Ni catalysts

Via elaborate synthesis control, self-supported Ni catalysts with specific structures could exhibit outstanding performances in SRM as well. By thermal decomposition of nickel tetra carbonyl, pure nickel powder with an open filamentary structure and irregular spiky surface was prepared by Rakass et al. [151] In their study, the $H_2$ production, $CH_4$ conversion as well as the reaction stability increased with the S/C ratio rising from 0.5 to 2.0. At the S/C ratio of 2.0, with the reaction temperature increasing from 300 to 700 °C, the $H_2$ production on the catalyst began at 325 °C and reached maximum at around 650 °C, while the $CH_4$ conversion increased constantly and was beyond 95% when the temperature was over 600 °C. Besides, the $CH_4$ conversion at 700 °C could kept stable for 100 h and no coke was formed, indicating the good stability and high coke resistance of the catalyst.

Honeycomb structured Ni has attracted a number of attention in the past few years (Fig. 4D) [96,95,97]. The Brunauer–Emmett–Teller (BET) surface area of the honeycomb-Ni catalyst (59.4 $cm^2/cm^3$) was nearly 30 times larger than commercial Ni based catalysts, which could be an important reason for its high activity. The selectivity to CO of honeycomb-type Ni was a little bit lower than that of the commercial Ni based catalyst, however, its activity was 3 times higher. The high coke resistance of the honeycomb Ni catalyst might be due to the exposure of the flat Ni (001) surface, which was not favorable for carbon deposition. However, the activity of honeycomb catalyst decreased in the first several hours due to the oxidation of surface Ni in the reaction.

Also impressively, a Ni coil catalyst with a high geometric surface area (88.1 $cm^2/cm^3$) was synthesized by Hirano et al. [152] This catalyst achieved 94% $CH_4$ conversion, 77.6% $H_2$ production, and 91.1% CO selectivity at 800 °C at an S/C ratio of 1.24.

## 3.2. Other non-noble metal catalysts

Besides Ni catalysts, other non-noble transition metal (e.g. Co, Cu, and Mo) based catalysts also show potentials for SRM. Some of them even showed higher catalytic activity than Ni based catalysts. For instance, $MoC_2/Al_2O_3$ exhibited significantly higher $CH_4$ conversion (95%) than Ni/$Al_2O_3$ (17.8%), even at a lower reaction temperature (700 °C vs. 750 °C) [153]. However, most of the catalysts suffer from low reducibility [32], easy deactivation (due to the oxidation of the metallic species), carbon deposition, and high-temperature induced aggregation [32,33,110,122,120]. Addressing these issues are the key for the wide application of these transition metal catalysts.

**1) Co Based Catalysts**

Co based catalysts exhibited promising performance toward SRM. However, deactivation of the Co catalysts due to oxidation of metallic Co by $H_2O$ was a critical issue. Adding noble metals on the Co/$Al_2O_3$ catalyst could ensure a more stable metallic state of Co and made the catalyst less susceptible to deactivation during SRM. The present of Pd, Pt, Ru, and Ir markedly reduced the reduction temperature of both $Co_3O_4$ and Co surface species via the hydrogen spillover effect [32]. 0.3 wt% Pt, Pd, and Ir all enhanced the $CH_4$ conversion on Co/$Al_2O_3$ from ~7% to 50–60% at 700 °C with the S/C ratio of 4.0, with Pt presenting the highest $CH_4$ conversion, $H_2$ production, and stability. The doping of non-noble transition metals could, to a certain extent, improve the activity of Co catalysts as well. The Co/Mg/Al catalyst promoted by La and Ce reached 85% $CH_4$ conversion at 700 °C and S/C ratio of 2.0 [154,155]. At a lower S/C ratio of 0.5, the unpromoted catalyst suffered from severe coke formation, whereas the La and Ce promoted catalysts were highly resistant to carbon deposition. However, a certain deactivation was still observed for the oxidation of Co. Promisingly, with the co-existence of Zr, La, and Pt with Co, Co–Pt–Zr–La/$Al_2O_3$ catalyst could achieve a nearly complete $CH_4$ conversion (99.3%) at 750 °C at an S/C ratio of 1.25 without any sintering or carbon deposition [110].

**2) Cu Based Catalyst**

An early study suggested pure Cu metal was relatively less active for SRM with the reaction steps being more uphill energetically, compared to metallic Pt, Pd, Rh, and Ni [156]. However, in the recent years, Cu has been proven an effective promoter for Ni catalysts [128] and $Co_6Al_2$ supported 5 wt% Cu catalyst showed promising SRM activity [33]. 5 wt % Cu/$Co_6Al_2$ could realize an almost complete convertion of $CH_4$ at 700 °C (S/C ratio of 3.0) without coke formation. However, increasing Cu content led to deactivation due to the agglomeration of copper oxide. Yet, higher Cu content resulted in higher selectivity to CO due to the enhanced reverse-WGS (RWGS).

**3) Mo Based Catalysts**

Mo could not only serve as a good promoter for Ni based catalyst [34], but also function as an active species for SRM. $Al_2O_3$ supported

$Mo_2C$ exhibited promising activity with ~95% $CH_4$ conversion rate and $H_2$ selectivity of ~70% at 700 °C and S/C ratio of 4.0 [153]. However, unsupported $Mo_2C$ showed a lower $CH_4$ conversion of ~40% and $H_2$ selectivity of ~35%. Such difference was probably due to the phase transfer from β-$Mo_2C$ in the unsupported $Mo_2C$ to α-$Mo_2C$ in the $Al_2O_3$ supported one.

### 3.3. Noble metal catalysts

Noble metal catalysts (e.g. Rh, Ru, Pt, Pd and Ir etc.) generally showed higher catalytic activity and stability than Ni catalysts [35]. However, their application was limited by their high cost. Meanwhile, some of them also suffer from aggregation and carbon deposition. Thereby, a number of research efforts have been paid to optimize the SRM performance of noble metal catalysts while decreasing their loading amount.

#### 3.3.1. Ru based catalysts

The activity and stability of Ru/$Al_2O_3$ catalyst have been improved by tuning the structure and chemical composition of substrate. $Al_2O_3$@Al core-shell structure consisting an Al metal core with high surface area $Al_2O_3$ shell was employed as the support for Ru [100]. The $Al_2O_3$@Al particles would aggregate during synthesis process, forming a secondary structure which was advantageous for an efficient heat transfer and a high dispersion of Ru. Accordingly, the Ru/$Al_2O_3$@Al exhibited higher activity than Ru/$Al_2O_3$ and no deactivation was observed during 40 h reaction. With La element forming a thin, homogeneous, and amorphous surface $La_2O_3$ layer on $Al_2O_3$, the Ru species in Ru/La–$Al_2O_3$ were detected as $RuO_2$, indicating an enhanced mater-substrate interaction [99]. Nearly complete $CH_4$ conversion (97.3%) and 78.3% $H_2$ yield could be realized at 800 °C on Ru/La–$Al_2O_3$ with a monolith support. When $Ni_6Al$ was used as the support and a small Ru loading amount (0.5 wt%) was applied, $RuO_2$ particles would be formed and disperse uniformly on the support surface [101]. A synergistic effect between Ru and Ni could occure and contribute to a high SRM activity. Ru/$Co_{6-x}Mg_xAl_2$ catalyst exhibited outstanding activity in SRM with a complete $CH_4$ conversion at 700 °C [102]. However, increasing Mg content could lead to the agglomeration of $RuO_2$ partilces and thus increased $RuO_2$ particle sizes. As a result, $CH_4$ conversion and $H_2$ yield increased with the decreasing of Mg content at low temperatures (Fig. 5A). The Ru/$Co_6Al_2$ exhibited the best activity and an excellent stability (Fig. 5B and C).

Despite of the negative effect of MgO in the composite oxides in some reports [101,102,157], both MgO and $Nb_2O_5$ supported Ru catalyst showed appreciable performance for SRM [103]. MgO supported Ru achieved a $CH_4$ conversion higher than 99%, a $CO_2$ selectivity of 62%, and a $H_2$ concentration of 78% in the outlet at 750 °C, while $Nb_2O_5$ support led to complete $CH_4$ conversion, 71% $CO_2$ selectivity, and 78% $H_2$ in outlet at the same temperature. The electron transfer from support to Ru metal particles might be the reason for high catalytic performance on Ru/MgO. The better catalytic performance of Ru/$Nb_2O_5$ could be attributed to two reasons: $Nb^{5+}$ ion on the surface could interact with Ru and the phase transfer of $Nb_2O_5$ from amorphous to tetragonal after the SRM reaction.

#### 3.3.2. Rh based catalysts

Rh was one of the most active catalysts for $CH_4$ adsorption, activation, and desorption and thus a potential catalyst for SRM. Onset for SRM on Rh/$Al_2O_3$ was slightly higher than 700 °C [36]. $CH_4$ conversion on Rh/$Al_2O_3$ reached ~78% at ~875 °C with CO selectivity of ~70%, which was comparable with Ru. With higher metal content and reaction temperature, $CH_4$ conversion on Rh was slightly higher than Ru (78% vs. 72%) at a low S/C ratio of 1.77. Comparing the SRM activity of catalysts with Ru, Rh, and Pt deposited over $CeO_2$ and $Al_2O_3$ carriers, 1.5% Rh/$CeO_2$ showed the highest conversion of methane [158].

Tuning the textual properties of the support and introducing a promoter are effective means in controlling the performance of Rh catalysts. Rh particles were distributed within a narrow range of 1–3 nm on $Al_2O_3$ synthesized by flame spray pyrolysis (FAl), whereas the Rh particle size ranged from 2 to 7 nm on Al particle suspension (SAl) [105]. With the Rh content lower than 1 wt%, Rh/SAl showed a higher $CH_4$ conversion than Rh/FAl (69% vs. 58%) while Rh/FAl exhibited a higher activity at 5 wt% Rh loading (62% vs. 65%). Hydroxyapatite (HAP) supported Rh (1 wt%) showed the highest $CH_4$ conversion of 77% at 650 °C without deactivation in 30 h [106]. Atomically dispersed Rh catalyst on $Al_2O_3$ was designed for SRM [159]. Promoters were required to avoid the aggregation of Rh during SRM at 760 °C. $Sm_2O_3$ and $CeO_2$ in Rh/x$Sm_2O_3$-y-$CeO_2$-$Al_2O_3$ catalyst could effectively improve the reaction rate and stabilize the structure of Rh particles [160]. With the addition of $CeO_2$ promoters, the formation and stabilization of atomically dispersed Rh metal species could be enhanced due to the SMSI, namely, the metal-O-Ce bonds [159,161,162]. Besides, carbon deposition was removed by the active oxygen in $CeO_2$.

Rh/$MgAl_2O_4$ showed a $CH_4$ conversion of ~41% at 850 °C [54]. Both the carbon deposition and metal sintering were rare on Rh/$MgAl_2O_4$. Theoretical results suggested the Rh–O bonds were the primary form of metal-support interaction in Rh/$MgAl_2O_4$, which could modify the electronic structure of Rh and limit its sintering. Also, dissociative adsorption of $H_2O$ was enhanced at interface which facilitated the SRM. Rh could be highly dispersed on Sr-substituted hexaaluminate surface [163]. Sr-substituted hexaaluminate was stable under high-temperature conditions without surface area loss and could prevent Rh particles from sintering [163]. This catalyst could convert 45% $CH_4$ and show 69% $H_2$ selectivity at 740 °C.

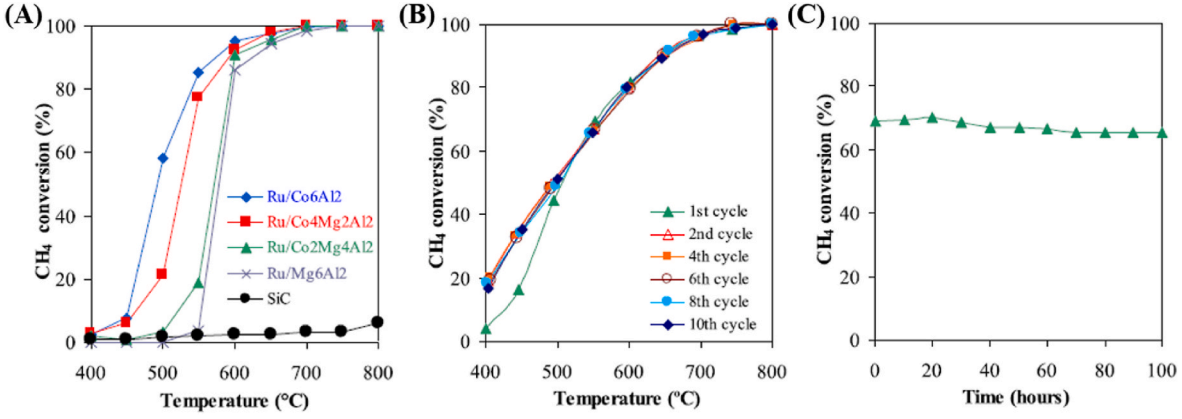



**Fig. 5.** $CH_4$ conversion on Ru/$Co_{6-x}Mg_xAl_2$ with different Mg content (A); Stability of Ru/$Co_6Al_2$ catalyst as a function of cycled times (B) and time (C). Reproduced with permission from Ref. [102]. Copyright 2014, Elsevier.

#### 3.3.3. Pt based catalysts

The performance of $Pt/Al_2O_3$ could be promoted by $CeO_2$, $La_2O_3$, and MgO. The $CH_4$ conversion on the Pt-NPs/$CeO_2$–$Al_2O_3$ with 3 nm Pt particle size was twice that on Pt-NPs/$Al_2O_3$ catalyst with similar Pt-NP sizes [37]. Bueno and co-workers studied $Pt/La_2O_3$–$Al_2O_3$ catalysts for SRM [164,165]. The addition of 12 wt% $La_2O_3$ led to the highest activity toward SRM. $La_2O_3$ could decrease the particle size of Pt, however, it could block active Pt sites as well. Also, $La_2O_3$ could narrow the electron density distribution on Pt sites and the formed $LaPtO_x$-like species could increase the $CH_4$ accessibility by promoting the removal of carbon deposition, indicating that SMSIs between Pt and $La_2O_3$ played an important role in promoting $Pt/La_2O_3$–$Al_2O_3$. The same group further promoted $Pt/La_2O_3$–$Al_2O_3$ by adding $CeO_2$ [108]. The addition of $CeO_2$ introduced $Ce^{4+}/Ce^{3+}$ redox couple, which could cooperate with $Pt^{\delta+}/Pt^0$ couple to enhance the activity and stability. Although the introduction of Ce decreased the dispersion of Pt, the carbon cleaning capacity was promoted via the high oxygen mobility on $CeO_2$, leading to a high stability of the catalyst. The optimal $CH_4$ conversion of 71.4% over 10 wt% $Pt/CeO_2$ only slightly dropped to 69.7% after 6 h running [109]. Also, with a small amount of MgO (Al/Mg = 5) in $Pt/Al_2O_3$, a $CH_4$ conversion (56%) 5 times that (10%) of the commercial nickel-based catalyst could be attained with 100% $H_2$ selectivity [166].

## 4. Advanced SRM processes

Despite its continuous progress, the conventional SRM process has several fundamental constraints. For example, it involves multiple steps and confronts severe operating conditions to obtain high conversions. To increase the $H_2$ selectivity, reduce the energy input or achieve better product separation, upgraded SRM processes, such as sorbent enhanced SRM (SESRM) [38], oxidative SRM (OSRM) [20,39,40], chemical looping process [41], photocatalytic SRM. solid oxide fuel cell (SOFC) [167–170]. steam plasma methane reforming (SPMR) [45], and electro-catalytic SRM [46] etc. have received increasing attention in the past years. The advantages and disadvantages of these processes are summarized in Table 2.

**Table 2**
Advantages and disadvantages of intensified SRM technologies.

| Intensified SRM processes | Advantages | Disadvantages |
|---|---|---|
| Sorbent enhanced SRM | High-purity hydrogen; Relatively low reaction temperature; Reduced energy consumption. | Relatively poor chemical stability. |
| Oxidative SRM | High $CH_4$ conversion; Reduced reaction temperature. | Low selectivity to $H_2$; Oxidation of the catalyst. |
| Chemical looping SRM | Reduced reaction temperature and energy consumption; Produce sequestration-ready $CO_2$ streams; Fuel flexibility; Higher heat-transfer coefficient. | Easy deactivation by carbon deposition; Low syngas selectivity; Require appropriate oxygen carriers. |
| Photocatalytic SRM | Environmentally friendly; low energy input. | Low conversion efficiencies; Limited reaction area. |
| Thermo-photo hybrid SRM | Environmentally friendly; Low reaction temperature; High solar energy utilization; Direct formation of $CO_2$ and $H_2$. | Limited reaction volume. |
| Solid oxide fuel cell | High efficiency; Design modularity. | Easy carbon deposition; Complicated system. |
| Plasma SRM | Variable product composition; Compactness; Fast response time; No catalysts needed | Reaction mechanism is not clear; Relatively poor controllability |
| Electrical field enhanced SRM | Low reaction temperature; Easy control | Relatively low selectivity to hydrogen |

### 4.1. Sorbent enhanced SRM

Since Rostrup-Nielsen first added $CO_2$ sorbent in SRM in 1868, sorbent enhanced SRM (SESRM), which integrates a $CO_2$ sorbent to the catalyst to drive the equilibrium shift of the reversible the SRM and WGS reactions toward hydrogen production (according to Le Chatelier's principle), has been a promising technology for high-purity hydrogen production [171]. The $CO_2$ sorbent in SESRM can be either physically mixed with the catalyst or combined with the catalyst to form a combined sorbent-catalyst material (CSCM) [172,173]. The hybrid catalyst-sorbent materials have attracted more attention (Table 3) since Strio's pioneering work in 2005, because they could not only eliminate mass diffusional limitation but also reduce the reactor volume [38,174,175].

The loading amount and properties of the catalyst species in a CSCM play an important role in determining the catalytic performance. Typically, Assabumrungrat and co-workers revealed that the promoting effect of the sorbent on the performance became obvious only when the Ni loading excessive a certain value (10 and 12.5 wt%) for Ni/CaO [38]. On an Al-stabilized CaO-nickel hybrid sorbent-catalyst, 25 wt% Ni led to the highest CaO conversion, $CH_4$ conversion (99.1%), and $H_2$ yield (96.1%) [180]. In the physically mixed system of NiO, CaO, and calcium cement aluminate (source of $Ca_{12}Al_{14}O_{33}$), García-Lario and co-workers received a ~93.5% $H_2$ in the outlet and 98% $CH_4$ conversion as well as a good stability after 7 cycles with 14% NiO in the composite [183]. When the NiO loading was decreased to 9%, only 75% $CH_4$ could be converted.

Meanwhile, different Ni precursors also had obvious influence on Ni-mayenite catalysts [187]. The acetate (Ac) Ni precursor resulted in NiO with a larger crystalline size. The carbonaceous matrix from acetate limited the accessibility of Ni by $CH_4$ but it didn't prevent Ni particle from sintering. In contrast, the NiO synthesized from nickel nitrate hexahydrate was with a smaller size and a higher surface area, allowing it to disperse evenly on the supports and achieve desirable $CH_4$ conversion in SESRM.

CaO is the most widely applied $CO_2$ acceptor in SESRM due to its high $CO_2$ adsorption capacity, excellent multicycle durability, and easy availability from natural dolomite and limestone [192,193]. However, one main drawback of CaO is its high regeneration temperature (up to 950 °C), which is energy consuming and could cause the sintering of the materials and thus dramatic performance decay after multiple cycles.

In 2010, Martavaltzi and Lemonidou developed a multifunctional NiO–CaO–$Ca_{12}Al_{14}O_{33}$ CSCM for SESRM [175]. Due to the synergistic effect of NiO and $Ca_{12}Al_{14}O_{33}$, the $CO_2$ fixation capacity of the material only showed a moderate loss (15%) after 45 sorption-desorption cycles. A rich $H_2$ (90%) stream was obtained at 650 °C with an S/C ratio of 3.4. Later, Müller et al. achieved the production of ultra-high-purity (99 vol %) $H_2$ in pre-breakthrough stage (before CaO saturation) by mixing Al stabilized CaO (CaO–$Ca_{12}Al_{14}O_{33}$) with a Ni-hydrotalcite-derived catalyst [182]. The well-dispersed $Ca_{12}Al_{14}O_{33}$ provided a stable nano-sized framework for CaO grain, leading to a high thermal stability of the Ca based pellets. Nevertheless, efforts are still needed to extend the period with ultimate-purity $H_2$ production during the process.

The free CaO/mayenite ratio and the morphology of the substrate are important for SESRM performance. When combining $Al_2O_3$ via wet mixing method, a proper amount of free CaO could facilitate the formation of $Ca_{12}Al_{14}O_{33}$ [188]. Unfavorable $CaAl_2O_4$ could form with a low CaO loading, while excessive free CaO could hinder the interactions between Ni and mayenite and deteriorate the catalytic activity. Gallucci et al. synthesized CaO–$Ca_{12}Al_{14}O_{33}$ with a low content (3.12 wt%) of Ni [184]. When the amount of $Ca_{12}Al_{14}O_{33}$ was over 70 wt%, the catalytic deactivation could be substantially suppressed, promoting $H_2$ content up to ~94.5% in the outlet with $CH_4$ conversion of ~89%. The peculiar cage-like crystal structure of $Ca_{12}Al_{14}O_{33}$ and the presence of "free" oxygen in its lattice played an important role during the process. A highly stable Ni–CaO-mayenite CSCM with an optimal Ni and CaO

**Table 3**
Representative catalysts for SESRM reported in the past decade.

| Catalyst | Metal Loading (wt %) | Sorbent | Temperature (K) | Pre-breakthrough time (h) | $CH_4$ conversion (%) | $H_2$ yield (%) | S/C ratio | Reference |
|---|---|---|---|---|---|---|---|---|
| Ni | 12.5 | CaO | | 2.7 | | | 3 | [38] |
| Ni/Ce–$Al_2O_3$ | 10.6 | CaO | 773 | 0.4 | 95.7 | 97.3 | 3 | [176] |
| Ni/$ZrO_2$ | 16.5 | CaO | 873 | 0.3 | 99 | 91.4 | 4 | [177] |
| Ni/$Al_2O_3$ | 15 | CaO–$Ca_5Al_6O_{14}$ | 923 | 1.3 | 99.4 | 87 | 2 | [178] |
| Ni/ @$Ca_5Al_6O_{14}$ | 15 | $Ca_5Al_6O_{14}$ | 873 | 0.5 | 94 | 92 | 3 | [179] |
| Ni–$Al_2O_3$ | 19 | CaO–$Ca_9Al_6O_{18}$ | 873 | 0.5 | 99.1 | 82 | 4 | [180] |
| Ni–Mg–Al | | CaO– $Ca_9Al_6O_{18}$ | 873 | 0.75 | 98 | 97 | 4.2 | [172] |
| Ni(Al:Mg)Ox | 47 | CaO–$Ca_{12}Al_{14}O_{33}$ | | | | 99 vol% | | [181] |
| Ni–Mg–Al | 47 | CaO–$Ca_{12}Al_{14}O_{33}$ | 773 | 0.1 | | 99 vol% | 4 | [182] |
| NiO/CaO | 18.5 | CaO–$Ca_{12}Al_{14}O_{33}$ | 873 | 1.3 | 98 | 90 vol% | 3 | [183] |
| C30IL | 3.12 | CaO–$Ca_{12}Al_{14}O_{33}$ | 863 | 0.16 | 89 | 74 | 3 | [184] |
| Ni/CaO–$Al_2O_3$ | 5.2 | CaO–$Ca_{12}Al_{14}O_{33}$ | 873 | | 97 | 90 vol% | 4 | [185] |
| Ni | | CaO–$Ca_{12}Al_{14}O_{33}$ | 903 | 1.7 | 95 | 92 | 3 | [186] |
| Ni | 10 | CaO-mayenite | 973 | | 96.8 | 97.9 vol% | 3 | [187, 188] |
| Ni | | $Ca_{0.5}Mg_{0.5}CO_3$ | 1073 | | 100 | >90 vol% | 2 | [189] |
| Rh/$Ce_aZr_{1-a}O_2$ | | $K_2CO_3$-hydrotalcite | 673 | | 99 | 99 | 5 | [190] |
| Ru | | CaO–$Ca_3Al_2O_6$ | 723 | 1.3 | 100 | 96 vol% | 4 | [191] |

fractions of 10 and 13.5 wt%, respectively, was developed by the same group via wet mixing and wet impregnation methods [194]. Although the $H_2$ yield was ~90 and ~77 vol% in the pre-breakthrough and post-breakthrough stages, respectively, the $CH_4$ conversion kept over ~95% in both stages. The excellent catalytic activity and $CO_2$ sorption capacity (~27 $g_{CO2}$/h per 100 g CSCM) could be maintained throughout 205 SESRM/regeneration cycles with the SESRM process carried out at 1 atm, 650 °C and the generation at 1 atm, 850 °C in $N_2$ (Fig. 6). Recently, Simonetti and coworkers suggested that CaO nanofibers with $Ca_{12}Al_{14}O_{33}$ as the spacer to prevent the sintering of CaO could maintain 94% of their initial performance after ten reforming-regeneration cycles [195].

Other Ca–Al oxide compounds such as CaO–$Ca_5Al_6O_{14}$ [178] and CaO–$Ca_9Al_6O_{18}$ [179,172,180] have been employed as effective supports for bifunctional CSCMs as well. Cheng and co-workers fabricated Ni/CaO–$Al_2O_3$ bifunctional catalysts containing NiO, CaO, and $Ca_5Al_6O_{14}$ by a sol-gel method [178,196]. The sample with a CaO/$Al_2O_3$ mass ratio of 6 developed by Cheng's group showed a stable performance for 50 cycles, indicating its promising stability [178]. The sol-gel

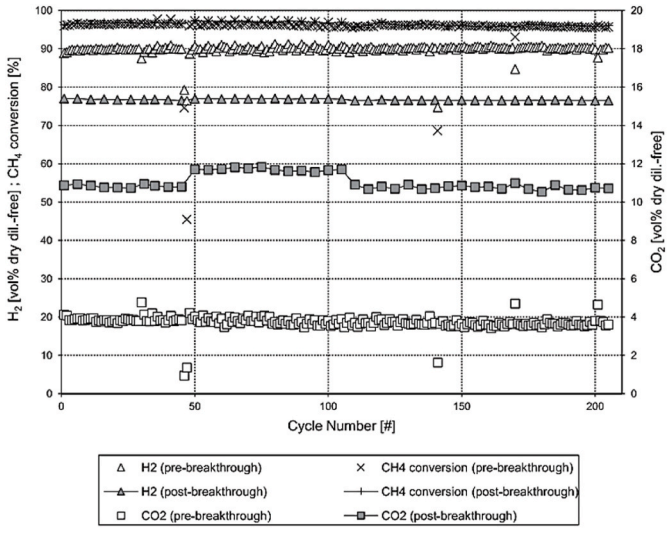



**Fig. 6.** Concentration of products and methane conversion detected before and after the breakthrough phase, as functions of the SESRM/regeneration cycle number. Reproduced with permission from Ref. [194]. Copyright 2018, Elsevier.

method was also employed by Zhou's group to prepare CaO–$Ca_9Al_6O_{18}$ for Ni based catalyst whose $CO_2$ capture capacity remained as high as 0.59 g $CO_2$/g sorbent at the 35th cycle, indicating that sol-gel method was effective to achieve high stability of sorbent [197]. Later, a CaO–$Ca_9Al_6O_{18}$@$Ca_5Al_6O_{14}$/Ni bifunctional catalyst with a CaO–$Ca_9Al_6O_{18}$ core and a Ni/$Ca_5Al_6O_{14}$ shell was designed by the same group (Fig. 7) [179]. $Ca_9Al_6O_{18}$ in this structure stabilized the structure by preventing the sintering of CaO and facilitating the diffusion of $CO_2$ through $CaCO_3$ layer. Meanwhile, $Ca_5Al_6O_{14}$ in the shell allowed a good dispersion of Ni particles. Thereby, high activity and stability over 60 cycles along with the maintained nearly complete utilization of CaO was demonstrated on the optimized sample with 13 wt% CaO and a core/-shell mass ratio of 0.2.

Apart from aluminates, several other metal oxides such as $La_2O_3$, [198] hydrotalcite [181], and $CaZrO_3$ [177] have also been explored in the past decade as stabilizer for CaO or CaO/$Al_2O_3$ in CSCM. A $La_2O_3$-modified NiO–CaO/$Al_2O_3$ complex catalyst was synthesized by Feng et al. via isometric impregnation [198]. The incorporation of $La_2O_3$ not only helped to maintain the support surface area and reduce the sintering of nano-$CaCO_3$, but also enhanced the interaction between NiO and the support, thus improving the stability of the catalyst from 7 up to 30 SESRM-regeneration cycles. Müller and co-workers synthesized a bifunctional catalyst with CaO and Ni particles highly dispersed on an $(Al:Mg)O_x$ matrix by the calcination of a hydrotalcite based precursor containing Al, Ca, Mg, and Ni [181]. In contrast to limestone, CaO supported by the porous and thermally stable Mg–Al oxide network showed superior $CO_2$ absorption and longer cycle life. During SESRM with this complex catalyst, 99 vol% hydrogen was formed in the effluent stream along with a good cyclic stability. The active CaO amount only reduced from 84% to 74% after 10 cycles. However, due to the low amount of CaO in the system, the $CO_2$ uptake was relatively low (0.074 g $CO_2$/g sorbent in average). Also, the incorporation of $CaZrO_3$ with CaO prevented CaO particles from sintering (Fig. 8) [177]. The CaO–Zr/Ni bifunctional sorbent-catalyst with an optimal Ni loading of 20.5 wt% maintained a $CH_4$ conversion of ~99% and an average $H_2$ yield of ~89% for 10 cycles with only a slight deactivation.

Furthermore, besides being a sorbent, CaO in Ni/CaO CSCM could also enhance the attrition resistance, sulfur tolerance, and coking resistance [185]. But studies in these aspects were relatively few. A study based on recent density functional theory (DFT) calculation clearly described the behavior of Ca species in preventing Ni from coke [199]. It was revealed that incorporated Ca in NiO could enhance $CH_x$ and $H_xO$ adsorption on the catalyst surface. As a result, the dissociation

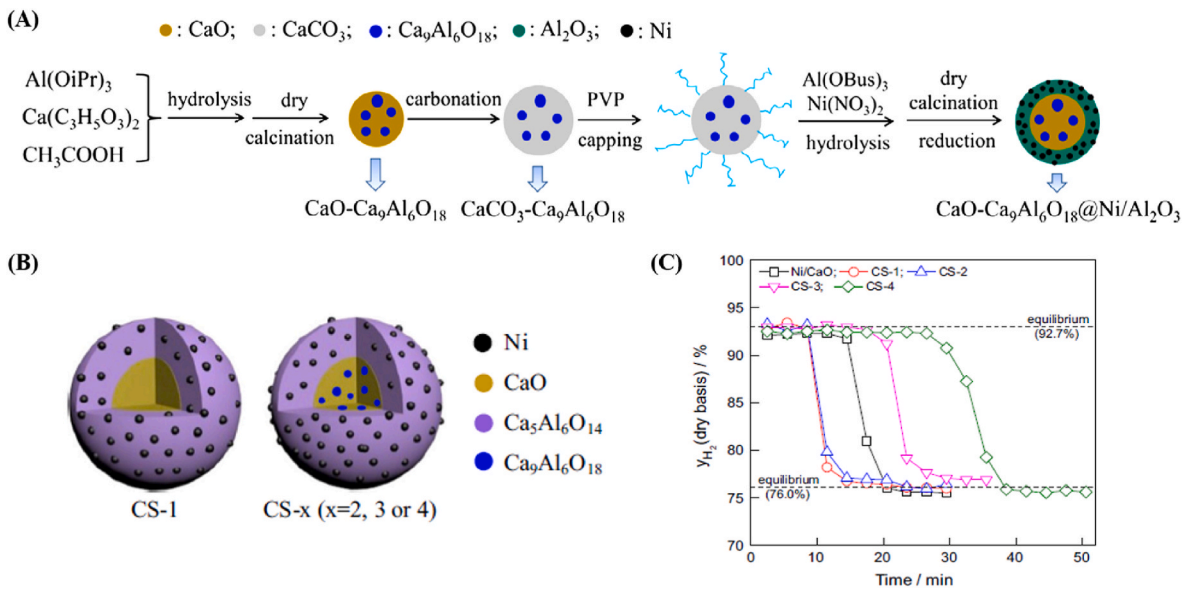



**Fig. 7.** (A) Preparation of the core-shell structured bifunctional material; (B) Actual configuration of core-shell structure; (C) $H_2$ production over different CaO–$Ca_9Al_6O_{18}$@$Ca_5Al_6O_{14}$/Ni catalysts at the first cycle. Reproduced with permission from Ref. [179]. Copyright 2017, Elsevier.

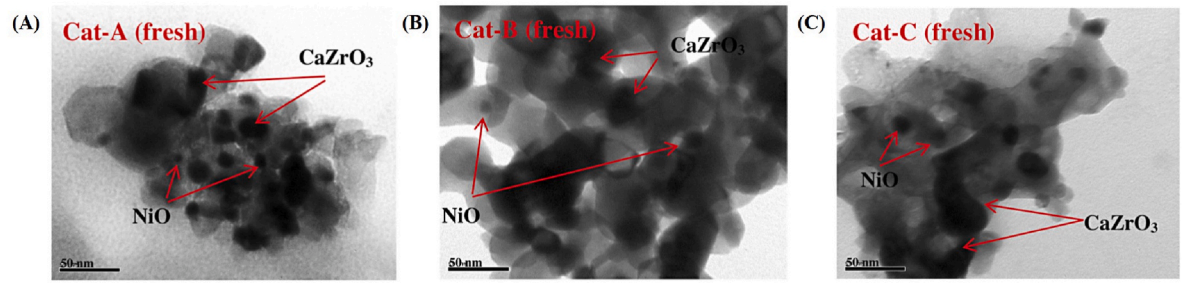



**Fig. 8.** TEM images of fresh Ni/$CaZrO_3$ samples. Reproduced with permission from Ref. [177]. Copyright 2014, Elsevier.

of $*CH_x$ and $*H_2O$, which were slow steps in SRM, were accelerated. Also, Ca enhanced the mobility of O atom on the surface and enhanced the CO formation.

Besides Ni based catalyst, noble metal based catalysts also show potential for SESRM. The Rh/$Ce_aZr_{1-a}O_2$ catalyst combined with $K_2CO_3$-promoted hydrotalcite demonstrated appreciable SESRM performance with >99% $CH_4$ conversion and >99% $H_2$ yield at 450 °C which could be maintained for 30 cycles [190]. Compared with Ni based catalyst, despite the expensive price of Rh, the enormous reduction of the reactor size, material loading, catalyst/sorbent ratio, and energy requirements are beneficial key factors for the success of Rh based catalyst. Ru/$Ca_3Al_2O_6$–CaO exhibited nearly complete $CH_4$ conversion and 96 mol% $H_2$ production which slightly deactivated after 10 cycles [191].

Promoters play an important role in CSCM as well. K is an effective promoter for CaO based CSCM. The K-hydrotalcite exhibited a fast adsorption of $CO_2$ with a high sorption capacity of 0.95 mol/kg while only 8% $CO_2$ capacity lost after the 1st cycle [200]. The addition of Ce and Zr resulted in strong basic sites for $CO_2$ adsorption on hydrotalcite-like material, and hence improved $H_2$ production [201]. Especially, the Ce-promoted sample exhibited extremely high adsorption capacity (1.41 mol $CO_2$/kg sorbent). Coking was suppressed by the high surface area and basicity. Thus, the $H_2$ yield reached 97.1 mol% and maintained for 13 cycles on Ce-promoted one. Silica addition improved the dispersibility of Ca sorbent and enabled rapid transfer of $CO_2$ to reactive pores, leading to faster $CO_2$ adsorption [202]. Interestingly, the individual application of $Ba_2TiO_4$ received poor performance whereas the cooperation of between $Ba_2TiO_4$ and CaO led to high $CO_2$ adsorption capacity and good kinetics of CaO as well as high $CH_4$ conversion at relatively low S/C ratios [203].

With the on-going development of the various catalysts for SESRM, increasing attention has been paid on their potential for long-term industrial-scale usage. Giuliano et al. examined the long-time (200 SESRM/regeneration cycles and >500 h non-stop per test) performance and mechanical stability of a 2-particle system and CSCM composed of Ni-catalyst and CaO–$Ca_{12}Al_{14}O_{33}$ [185]. With reforming at 650 °C and oxidative regeneration under pure $CO_2$ at 925 °C, both systems exhibited satisfactory catalytic stability and good resistance to attrition. The $H_2$ yield of both systems could reach the thermodynamic limits in both pre-breakthrough (96.9 vol%) and post-breakthrough (77.4 vol%) stages. The $CH_4$ conversion on CSCM was maintained at slightly higher value (~94%) than that in the 2-particle system (~93%). However, a certain decrease in sorption capacity was observed due to the change of their physical and chemical properties during the tests. Due to the coexistence of the three species (Ni, CaO, and $Ca_{12}Al_{14}O_{33}$) in the same particle, it would be more challenging to keep the chemical stability of CSCM than that of the segregated particle system. Thereby, simple and highly stable supports with strong interactions with both the catalyst(s) and sorbent(s) should be endeavored in future research to obtain CSCM with industry-admirable durability.

### 4.2. Oxidative SRM

Due to the endothermic nature of SRM process, external heat must be supplied. To reduce the energy input in SRM process, partial oxidation of

methane (POX), which is exothermal, has been combined with SRM by adding a small amount of oxygen into the SRM reaction system since 2002 [114,204]. This combined process, which is called oxidative SRM (OSRM) or autothermal reforming process (ATR), shows potential for high $CH_4$ conversion at low temperatures. One of the great challenges for OSRM was to prohibit the oxidation of catalyst [205].

The addition of noble metal has been extensively studied to improve the activity and stability of Ni catalysts for OSRM previously (Fig. 9) [206]. Recent efforts in OSRM mainly focused on enhancing the performance of Ni based catalyst via the selection of suitable bases and developing noble metal based catalyst for advanced performance. $Ni/NiAl_2O_4/Al_2O_3$ spinel-derived Ni catalyst was constructed to attain a $CH_4$ conversion of ~90% at 600 °C in OSRM without coking [114]. The addition of $CeO_2$ and $La_2O_3$ to $Al_2O_3$ exhibited limited improvement to $CH_4$ conversion rate but improved the stability and coking resistance of $Ni/Al_2O_3$ catalyst [207,208].

Among the various noble metal catalysts, $Rh/CeO_2$ showed the best performance for OSRM with a $CH_4$ conversion of 98.3%, a $H_2$ concentration of 67.9% in the outlet, and a $H_2/CO$ mole ratio of 3.2 [115]. $CeO_2$–$ZrO_2$ prepared by micro emulsion was found to exhibit better performance as the support for Rh catalyst in OSRM (~75% $CH_4$ conversion at 750 °C) than the coprecipitation prepared one (~60% under the same condition), due to a higher oxygen exchange capacity and a lager pore diameter and allowed a better dispersion of Rh [209]. Besides, the coprecipitation prepared sample was deactivated by carbon deposition, while no carbon was observed on micro emulsion prepared one.

However, different from SESRM, the selectivity to $H_2$ in OSRM was low and the major product was carbon oxide. This great disadvantage has limited its application for syngas gas and hydrogen production.

### 4.3. Chemical looping SRM

Chemical looping SRM (CLSRM), which could generate pure hydrogen and syngas separately or simultaneously, has received great attention in the recent two decades [210,211]. As shown in Fig. 10, the system for CLSRM generally consists of a fuel reactor (FR) and a steam reactor (SR) [41]. In the FR, the oxygen carrier, i.e. the catalyst, is reduced by the fuels, and the fuels are oxidized to generate $CO_2$, CO, $H_2$, and $H_2O$, while in the SR, $H_2O$ is reduced by the oxygen carrier to $H_2$. Usually, an air reactor (AR) is added to confirm the complete recovery of the oxygen carrier, as shown in Fig. 10b. Pure hydrogen can be obtained from the outlet of SR by simply cooling and condensing the steam, without any additional gas treatments required.

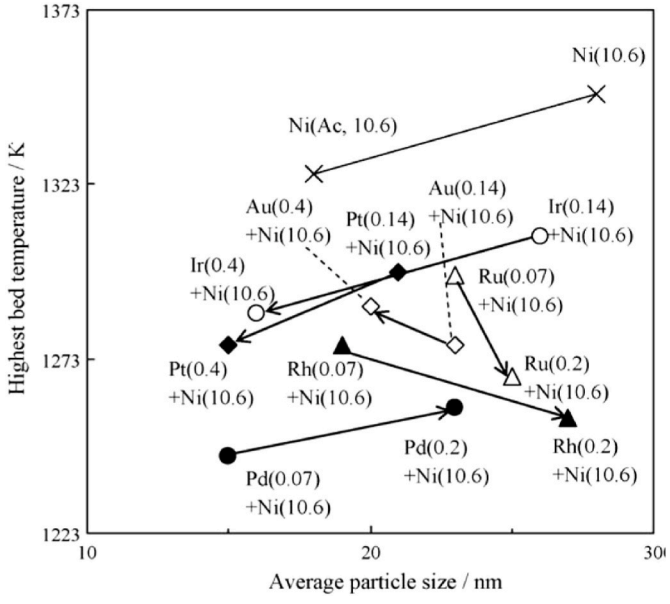



**Fig. 9.** Relation between average particle size and highest bed temperature in oxidative steam reforming. Reproduced with permission from Ref. [206]. Copyright 2009, Elsevier.

Studies of CLSRM process have mainly focused on oxygen carrier screening and performance optimization. A proper oxygen carrier material needs to have a sufficiently high reactivity toward methane activation to produce syngas, a good performance for water splitting to produce hydrogen, and a high stability during redox cycles. Besides, carbon deposition resistance is also needed for the oxygen carrier [41]. Because if the particles in FR was deposited by carbon, the surface carbon will react with steam in SR and contaminate the $H_2$ generated.

Up to now, a series of metal oxides have been considered as possible oxygen carriers for CLSRM, such as $Fe_3O_4$ [212], $WO_3$ [213,214], $SnO_2$ [215], Ni-ferrites [216], (Zn, Mn)-ferrites [217], Cu-ferrites [218–220], and Ce based oxides [221–226]. Some representative oxygen carries studied are listed in Table 4. Ni based oxygen carrier was demonstrated as one of the most promising oxygen carrier for CLSRM process due to its high activity and selectivity [41]. $NiO/SiO_2$ was an effective oxygen carrier for CLSRM with a high activity (~100% $CH_4$ conversion) and selectivity (72.7% $H_2$ selectivity) [227]. Different inert supports and preparation methods influenced the properties of Ni based oxygen carriers. For example, $NiO/MgAl_2O_4$ exhibited higher activity, selectivity, and coking resistance than $NiO/NiAl_2O_4$ [227]. The Ni based oxygen carrier impregnated on $\gamma$-$Al_2O_3$ showed the lower reactivity than that on $\alpha$-$Al_2O_3$ which was due to the solid state reaction between the NiO and the $\gamma$-$Al_2O_3$ to form $NiAl_2O_4$. Sample prepared by dry impregnation method showed higher coking resistance than deposition-precipitation method prepared one, but the reason for that remained to be explained [228].

The coke resistance of Ni based oxygen carrier could be further enhanced by adding a small amount of alkali earth metals [245] or alkali earth metals oxide [246,247] or a trace amount of steam [247]. Recently, Bukur and co-workers comparatively studied the effect of $Al_2O_3$ and $ZrO_2$ as supports for NiO oxygen carrier [231,248]. $NiO/ZrO_2$ could maintain ~82% $CH_4$ conversion at 650 °C and an S/C ratio of 3.0 for 20 cycles. In contrast, $NiO/Al_2O_3$ gradually deactivated in the following cycles, although the $CH_4$ conversion rate was comparable in the 1st cycle. Meanwhile, carbon deposition was less on $NiO/ZrO_2$, indicating the higher coking resistance. The $Al_2O_3$-supported Ni–Fe oxide has drawn considerable attention as the oxygen carrier for CLSRM since it achieved $CH_4$ conversion up to 97.5% and CO selectivity up to 92.9% at 900 °C [230]. The selectivity to CO was enhanced due to the C–O bond association between the hot steam and the C from the selective cracking of $CH_4$ on Ni. The metallic Fe in the catalyst was active toward steam to produce $H_2$ and the $H_2$ production capacity showed a positive correlation with the Fe content, though, excessive Fe could result in an inferior $CH_4$ conversion [229]. Meanwhile, the Ni–Fe interaction could enable the lattice oxygen to be recovered by Ni under steam atmosphere. Moreover, the $Al_2O_3$ base was effective in preventing collapse of the spatial structure and the sintering of Ni–Fe oxide.

Ce or Y promoted Ni/SBA-16 oxygen carriers were developed by Rahimpour and co-workers for CLSRM recently [232,233,249,250]. The Ce-promoted 20Ni-11.6Ce/SBA-16 oxygen carrier had the high activity of about 100% $CH_4$ conversion and ~87% $H_2$ production yield at 700 °C while Y-promoted 25Ni-2.5Y/SBA-16 achieved comparable $CH_4$ conversion and $H_2$ yield at 650 °C. Both Ce and Y could promote the dispersion of NiO particle by reducing the particle size and suppressing the aggregation. Meanwhile, coking was eliminated due to the smaller NiO particle size at the presence of Ce and higher oxygen mobility at the presence of Y. A fibrous aluminosilicate support for NiO oxygen carrier was tested for CLSRM [244]. The application of fibrous aluminosilicate support increased the $CH_4$ conversion by up to 10% and $H_2$ yield by up to 5% which was due to the high malleability, high thermal stability, and large surface area.

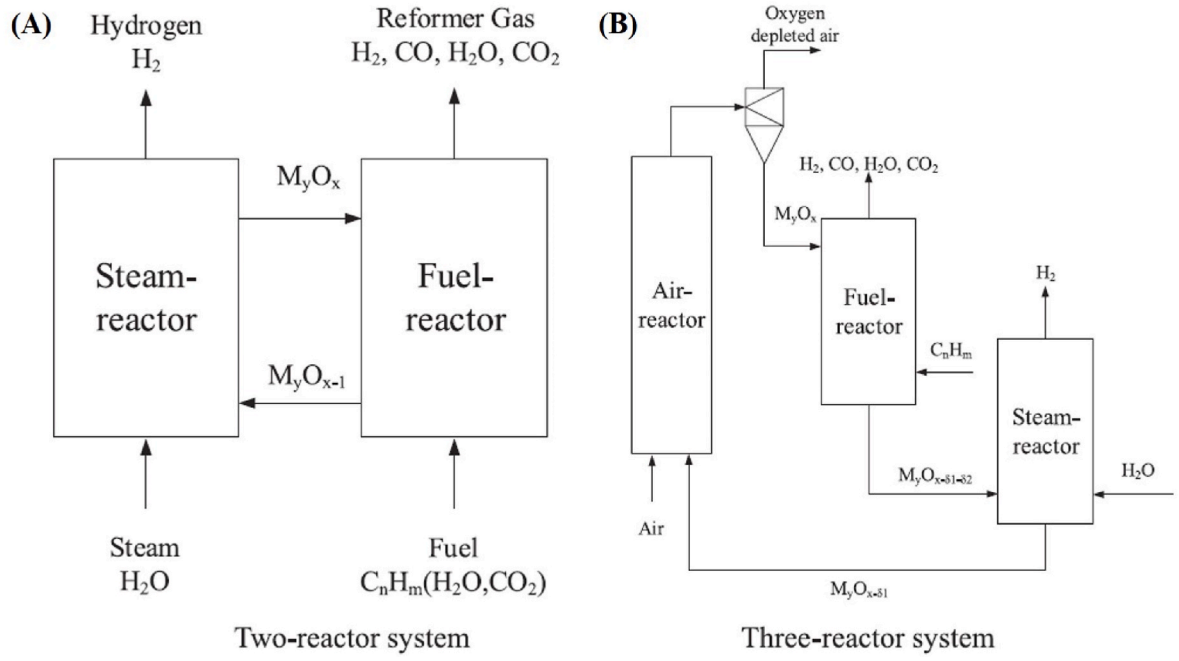



**Fig. 10.** The schematic diagram of (a) a two-reactor and (b) a three-reactor CLSRM system. Reproduced with permission from Ref. [41]. Copyright 2018, Elsevier.

**Table 4**
Representative oxygen carriers for CLSRM reported in recent years.

| Oxygen Carrier (OC) | Doping | Temperature (K) | $CH_4$ conversion (%) | $H_2$ selectivity (%) | Reference |
|---|---|---|---|---|---|
| $NiFe_2O_4$+$Al_2O_3$ | | 1123 | | 400 ml $H_2$ per gram Oxide Carrier | [229] |
| NiFeAl | | 1173 | 97.5 | 92.9 | [230] |
| NiO/$Al_2O_3$ | | 873 | 78 | ~70 | [231] |
| NiO/$ZrO_2$ | | | 82 | ~75 | |
| NiO/SBA-16 | Ce | 923 | 99.6 | 84.3 | [232] |
| NiO/SBA-16 | Y | 873 | 99.8 | 85.3 | [233] |
| Fe/$Al_2O_3$ | Mg | 873 | 100 | 83.7 | [234] |
| $Fe_2O_3$/$CeO_2$ | | 1073 | ~43 | ~32 | [235] |
| $LaFe_{0.7}Mn_{0.3}O_3$ | | 1073 | 97.3 | | [236] |
| $LaSrFeCoO_6$ | | 1073 | 78 | 44 | [237] |
| $La_{1.6}Sr_{0.4}FeCoO_6$ | | 1073 | ~90 | 93.8 | [238] |
| $La_{0.7}Sr_{0.3}FeO_3$ | | 1073 | 80 | 96 | [239] |
| $Ba_{0.3}Sr_{0.7}CoO_{3-\delta}$/$CeO_2$ | | 1073 | | ~85 | [240] |
| CaCoZr | | 923 | 98.3 | 84.5 | [241] |
| CaCoY | | 923 | 95.8 | 82.9 | [242] |
| $CeO_2$–$ZrO_2$ | | 1023 | 55.9 | 89.1 | [243] |
| Ni–Co/aluminosilicate | | 923 | 96 | 75.9 | [244] |

The development of Fe based oxygen carrier was also a hot topic for CLSRM due to its abundance, high melting temperature, and low price [41]. Also, Fe based oxygen carrier exhibited high sintering resistance and capacity for oxygen adsorption [251]. However, the relative low reaction rate, low oxygen transport capacity, and low selectivity to syngas restricted the application of Fe based oxygen carrier [41]. The performance of Fe based oxygen carriers, previously promoted by $Rh_2O_3$–$Y_2O_3$, $Cr_2O_3$–MgO [252], and CuO [253], were proved to be selective to syngas while the addition of Mg [234], Ca [254,255], or Ce [255] led to high $H_2$ production capacity and stability. In 2011, Ortiz and co-workers investigated a Fe based waste product as the oxygen carrier for CLSRM. The high oxygen transport capacity led to ~60% $CH_4$ conversion [256]. Recently, a $Fe_2O_3$/$Al_2O_3$ oxygen carrier prepared by impregnation method was reported with high sintering and coking resistance [257]. This oxygen carrier could reach a complete $CH_4$ conversion at 880 °C and an S/C ratio higher than 1.5. A Ce–Fe mixed oxides exhibited ~45% $CH_4$ conversion at 900 °C [235]. The formation of $CeFeO_3$ enhanced the reducibility of the oxygen carrier.

Perovskite oxygen carriers serve as both the lattice oxygen carriers and catalysts for $CH_4$ activation. Generally, perovskite oxygen carrier was advantageous in oxygen mobility, thermal stability, and selectivity to syngas. La–Fe-contained perovskite was reported as good oxygen carrier for $CH_4$ conversion to syngas [258–261]. In the recent decade, Zhao et al. did a series of studies on La–Fe-contained perovskite oxygen carrier [236,262]. The effect of Mn and Ni substitution of $LaFe_{1-x}M_xO_3$ (M = Mn or Ni) on the oxidation activity and coking resistance in CLSRM was investigated. The substitution of mixed $Mn^{4+}$ and $Mn^{3+}$ led to the mixed states of $Fe^{4+}$, $Fe^{3+}$, and $Fe^{2+}$. The mixed oxidation state not only increased the lattice oxygen, but also enhanced the mobility of lattice oxygen from bulk to surface which could enhance the reactivity toward $CH_4$. The value of x in the range of 0.3–0.5 was appropriate while higher Mn content led to methane decomposition and low $H_2$ generation. The Ni-substitution also led to mixed oxidation states of Fe. Meanwhile, Ni substitution was desirable for more adsorbed oxygen and restraining the formation of carbon deposition. However, high Ni content led to easier carbon deposition due to enhanced $CH_4$ dissociation. Thus, the value of $x = 0.1$ was appropriate for Ni substitution. $LaSrFeCoO_6$ double perovskite exhibited ~80% $CH_4$ conversion at 850 °C and stable performance

for 10 cycles [237]. Fe and Co take part in the reaction, exhibiting mixed oxidation states. Among four tested preparation methods, micro-emulsion method was judged as the best preparation method. The sample oxidized by steam exhibited higher $CH_4$ conversion than air [238]. The Sr substitution induced the Fe/Co disorder generating oxygen vacancies and/or higher oxidation state of metal ions which could strongly enhance the reducibility of oxygen carrier [263]. $La_{1.6}Sr_{0.4}FeCoO_6$ exhibited the best performance with ~90% $CH_4$ conversion. The C–O bond association between *C from *$CH_4$ dissociation and lattice oxygen were the origins for syngas production. While the deep reduced metals ($Fe^{2+}$ and $Co^0$) combining with oxygen vacancies provided active site for $H_2O$ dissociation to generate $H_2$ [229]. Sr substitution also led to better catalytic performance on $La_{0.7}Sr_{0.3}FeO_3$ and $Ba_{0.3}Sr_{0.7}CoO_{3-\delta}/CeO_2$ oxygen carriers [239,240].

In the recent years, sorption enhanced chemical looping methane steam reforming (SE-CLSRM), which was first proposed by Lyon and Cole in 2000, has drawn increasing attention [264]. The exothermic sorbent carbonation, namely the CaO carbonation, provided heat for the endothermic reforming reaction, while the regeneration of CaO was driven by the oxidation of oxygen carrier [265,266]. Therefore, compared to general CLSRM, SE-CLSRM could realize a higher $CH_4$ conversion and $H_2$ production at a lower temperature (~650 °C) [267]. However, among $O_2$, air, $H_2O$, and $CO_2$, only pure $O_2$ could regenerate NiO oxygen carriers in SE-CLSRM [268]. Zr promoted Ca–Co based bifunctional catalyst-sorbent exhibited 98.3% $CH_4$ conversion and 84.5% $H_2$ yield at 700 °C [241]. Besides, the performance could be maintained in 16 cycles, while non-promoted Ca–Co deactivated after only 10 cycles. The addition of Zr not only enlarged the surface area and porosity of the oxygen carrier, but also suppressed the sintering of CaO particles. Y-promoted Ca–Co based bifunctional catalyst-sorbent also demonstrated appreciable performance with 95.8% $CH_4$ conversion and 82.9% $H_2$ yield at 700 °C for 16 cycles [269]. The addition not only improved the surface area and porosity of the catalyst, but also facilitated the regeneration of Ca–Co based bifunctional catalyst-sorbent. Ce-promoted Ca–Co based on $SiO_2$ support reached a nearly complete $CH_4$ conversion and produced 95% purity $H_2$ at 550 °C [270]. The ratio of $CeO_2$ to CaO in this catalyst-absorbent system is critical, since while $CeO_2$ improved the surface area, porosity and dispersion of the CaO sorbent, it reduced the amount of CaO available for reacting with $CO_2$. The Ca/Ce ratio of 0.94/0.06 was found to show the highest activity and sorption capacity. At a relative low reaction temperature (550 °C), $CeO_2$–$ZrO_2$ oxygen carrier exhibited the highest $CH_4$ conversion (~56%) with a $H_2$ selectivity of ~89%, which could be attributed to its uniform and 3D macroporous structure [243]. However, this unique structure collapsed after 10 cycles, leading to the low $H_2$ production.

### 4.4. Photocatalytic and thermo-photo hybrid SRM

Solar energy is an abundant energy source which is difficult to be captured and effectively utilized [271,272]. Photocatalytic SRM (PSRM) [42,273] and thermo-photo hybrid SRM (TPSRM) [43] are effective and environmental-friendly methods for capturing solar energy and converting it into chemical energy.

Yoshida's group reviewed a series of metal-loaded semiconductor photocatalysts for PSRM, including Pt-loaded $TiO_2$ (Pt/$TiO_2$), Pt-loaded La-doped $NaTaO_3$ (Pt/$NaTaO_3$:La), Pt-loaded $CaTiO_3$ (Pt/$CaTiO_3$), Pt-loaded $Ga_2O_3$ (Pt/$Ga_2O_3$), and Rh-loaded $K_2Ti_6O_{13}$ (Rh/$K_2Ti_6O_{13}$) [274]. Among them, Pt/$NaTaO_3$:La photocatalyst showed the highest hydrogen production rate (1.8 μmol $min^{-1}$) in their testing condition (300 W xenon lamp, 1.5 %$H_2O$/50%$CH_4$/Ar as the reaction gas with a flow rate of 50 ml $min^{-1}$). Furthermore, using a flux method, the group obtained $NaTaO_3$:La with cubic and rectangular morphologies, which demonstrated much higher PSRM activity than the sample prepared without the flux [275]. Besides, the PSRM performance of Pt/$NaTaO_3$:La could be further improved by increasing its crystallite size. Rh/$Na_2Ti_6O_{13}$ (NTO) prepared by a flux method showed the same relationship between crystal size and performance, with its hexagonal rod-like structure exhibiting the best performance [276]. Recently, PSRM have been carried out on Ru/γ-$Al_2O_3$, Rh/$TiO_2$, and La/$TiO_2$ at near ambient temperatures [277–279]. Also, Hong and co-workers reported a photocatalytic CLSRM [280]. However, the efficiencies of these conventional photocatalytic processes are far away from satisfaction for industrial application.

In contrast, Hu and co-workers developed a thermo-photo hybrid system for efficient SRM [43]. More impressively, the synergistic effect of the thermal and solar energies allowed the direct conversion of $H_2O$ and $CH_4$ into $H_2$ and $CO_2$, which requires two steps in traditional thermocatalytic processes (Eq. 1-2). This in fact solved the conflict between the high temperature required for $CH_4$ activation and the relatively low temperature needed for the production of $H_2$ and $CO_2$ immediately, and opened up a new opportunity for hydrogen production from water and natural gas using solar energy.

The reaction was successfully realized on a Pt/black $TiO_2$ catalyst dispersed on a $SiO_2$ with light-diffuse-reflection surface [43]. The black $TiO_2$ exhibited a narrow bandgap of about 1.0 eV (Fig. 11A) and thus was responsive to the visible light and even the near-IR light. Meanwhile, the light-diffuse-reflection-surface of $SiO_2$ substrate could increase the light absorption by 100 times. Without illumination, the catalyst was not active for SRM until the temperature reached to over 500 °C (Fig. 11B) [43]. In contrast, when the Pt/black $TiO_2$ catalyst was irradiated by either simulated AM 1.5G sunlight (Fig. 11C) or > 420 nm visible light (Fig. 11D), the synergy between the photo and thermal energies allowed the reaction to start at 200 °C, which is 300 °C lower than that without illumination (Fig. 11C and D). Meanwhile, compared to the general photocatalysis, the kinetic energies of the reactants was enhanced in the thermo-photo hybrid catalytic process [43]. Besides, it was found that the band gap of the semiconductor slightly and linearly decreased with the increasing temperature. As a result, an extraordinarily high apparent quantum efficiency of 60% was attained at 500 °C, and the $H_2$ yield was 3 orders of magnitude larger than the previous reports [43].

### 4.5. Methane fueled solid oxide fuel cells

Solid oxide fuel cells (SOFCs), which convert chemical energies (hydrogen and hydrocarbons) into electricity, are one of the most promising energy conversion systems due to its high efficiency, low pollution, and fuel flexibility [281]. In methane fueled SOFCs (i.e. internal reforming SOFCs), the waste heat from the electrochemical reactions and the joule heat can be used to supply the energy for the endothermic SRM reaction and the need of air-cooling for the cathode side can be reduced as well [44,282].

Nickel/yttria-stabilized zirconia (Ni/YSZ) cermet is the most commonly used anode catalyst for internal reforming SOFCs due to its high activity for electrochemical fuel oxidation reaction and compatibility of thermal expansion [281–283]. However, the serious coke formation and poor impurity (e.g. sulfur) tolerance of the catalyst significantly hindered its development [281,284]. A high S/C ratio (typically over 2) can to some extent suppress the carbon deposition, but this would reduce the conversion efficiency. Besides, it was recently found that Ni-YSZ catalyst showed fast degradation in complete internal steam reforming of methane at 750–950 °C, which could be caused by the depletion of Ni-catalyst via the formation of volatile $Ni(OH)_2$ at the fuel entrance [285].

The issues of Ni/YSZ can be mitigated by introducing $CeO_2$ or BaO in YSZ solid solution [286,287]. The $CeO_2$ particles on the surface of $Y_2O_3$–$CeO_2$–$ZrO_2$ anode effectively promoted the electrochemical activity in the SOFC by facilitating the migration of $O^{2-}$ to the active site of the Ni catalyst while suppressing the coke deposition [286]. Chen and co-workers reported the stable production of hydrogen through SRM on Ni–$BaZr_{0.8}Y_{0.2}O_3$(Ni-BZY) anode at a temperature as low as 550 °C with negligible coke, indicating the good catalytic activity and high coking

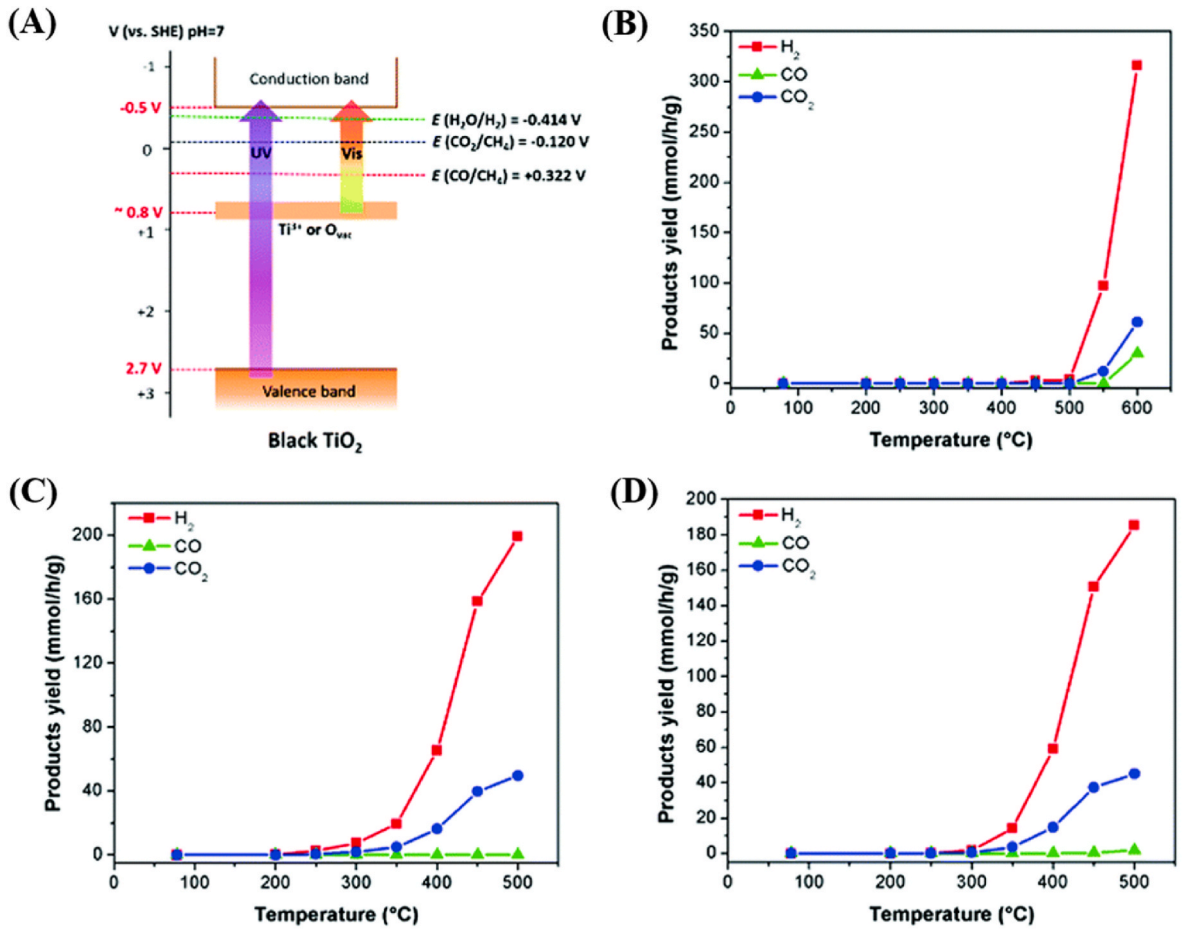



**Fig. 11.** (A) Relationship between the band structure of black $TiO_2$ and redox potentials of methane steam reforming. Product yields from steam reforming of methane ($H_2O/CH_4$ molar ratio 1:1 and GHSV = 80,000 ml $h^{-1}$ $g^{-1}$) over the Pt/Black $TiO_2$ catalyst dispersed on the light-diffuse-reflection-surface of a $SiO_2$ substrate: (B) without light irradiation; (C) under 1.5G sun light irradiation; and (D) under $\lambda > 420$ nm visible light irradiation. Reproduced with permission from Ref. [43]. Copyright 2019, Royal Society of Chemistry.

resistance of the anode materials [287]. However, when rising the temperature to 650 °C and 700 °C, serious coking took place at S/C ratio of 1.0.

Due to its high mixed ionic-electronic conductivity and ability to suppress the carbon formation, doped-ceria is frequently used to enhance the anode performance [288,289]. The doped species shed important influence on the performance. For instance, during SRM at 800 °C with an S/C ratio of 1.5, Ni–$Ce_{0.75}Zr_{0.25}O_2$ (Ni-CZO) stabilized in 24 h with a stabilized methane conversion of ~45%, while Ni–$Ce_{0.9}Gd_{0.1}O_2$ (Ni–CGO) needed 40 h to stabilize to ~38%. This could be ascribed to the higher amount of ex-solved Ni nanoparticles, lower $Ce^{4+}/Ce^{3+}$ reduction energy, as well as the better conductivity, thermal stability, $H_2/CO$ adsorption capacity, and compatibility with the YSZ-based electrolyte of the CZO support [290]. In addition, introducing a second dopant may help vary the coke deposition and sintering behaviors of the catalysts. The addition of Au [291] and $Fe_2O_3$ [292] on Cd-doped ceria has proven effective in enhancing the carbon tolerance and inhibiting the sintering of Ni catalyst, respectively.

Also, scandia-stabilized zirconia (SSZ) is a commonly used SOFC electrolyte due to its high stability below 700 °C and high ionic conductivity. Ni-SSZ showed appreciable performance for methane-driven SOFCs and the incorporation of Cu on Ni-SSZ can further enhance the coke resistance of the catalyst [293].

The n = 2 Ruddlesden-Popper (RP) phases like $La_{1.5}Sr_{1.5}Mn_{1.5}Ni_{0.5}O_{7\pm\delta}$ have been used to fabricate Ni-cermet catalysts with strong metal-support interaction. With a mixture of 82 mol %$CH_4$, 18 mol %$N_2$ and a low S/C of 0.15, Ni/$LaSrMnO_4$ exhibited 97% selectivity for CO production with 14.60 mol % $CH_4$ conversion and around 24.19 mol % $H_2$ production for more than 4 h [294].

Metals or metal oxides other than Ni have also been applied as active catalysts. For example, the $La_{0.8}Sr_{0.2}CrO_3$ (LSC) based Ru catalyst exhibited a good stability and a coking resistance under water deficient condition due to the interaction between Ru and LSC [295]. Since perovskite compounds of general formula $ABO_3$, such as $La^{3+}$ or $Y^{3+}$ doped-$SrTiO_3$ or $BaTiO_3$, show high resistance to reducing and sulfur-containing atmospheres, they have received a large amount of attention as alternatives to Ni-cermets [296,297]. But their electrochemical activity were relatively poor and needed to be enhanced by doping cations (e.g. $Ce^{n+}$, $Mn^{n+}$, $Ga^{3+}$) at the B site of the perovskite [298]. Especially, the promoting effect of Ce doping was significant: 20 times enhancement in activity can be achieved with 5 at% Ce substituted for Ti in $La_{0.05}Ba_{0.95}Ti_{0.9875}O_3$ [297].

Similarly, fluorite-type mixed oxides with general formula of $Ce_{1-x}A_xO_{2-\delta}$ (A = Pr, Sm, Gd; $0.1 \leq x \leq 0.2$) have been explored as SOFC anode materials for SRM under water deficient conditions (50% $CH_4$ and 5% $H_2O$ in $N_2$ balance) [299,300]. Among them, the $Ce_{0.8}Pr_{0.2}O_{2-\delta}$ sample exhibited the best catalytic performance with 1.46% $H_2$, 0.145% CO, and 0.26% $CO_2$ in the outlet gas at 840 °C [299]. This is because that a relatively higher amount of easily reducible and well dispersed praseodymium species ($Pr^{4+}/Pr^{3+}$) were interacting strongly with ceria ($Ce^{4+}/Ce^{3+}$) in this samples. Besides, it was found by the group that the presence of $H_2S$ in the feed was beneficial to the catalytic activity [300].

In addition, to improve the cell power output and suppress coke

formation over the anode, indirect methane-fueled SOFCs, which combine the SRM reaction and the oxidation of syngas on the anode, have been explored by loading active SRM catalysts (e.g. Ru–$CeO_2$) on the anode surface to increase [301]. The focus of studies in this area is to obtain cheap alternatives to Ru–$CeO_2$. Shao and co-workers suggested Ni–$Al_2O_3$ exhibited a comparable activity (100% $CH_4$ conversion at 750 °C) with Ru–$CeO_2$, and the coke formation on the anode was not obvious in the presence of water steam [302]. The maximum peak power densities reached 494 mW $cm^{-2}$ at 850 °C, which was comparable to those operating on hydrogen. By introducing lanthanide promoters, including $La_2O_3$, $CeO_2$, $Pr_2O_3$, $Sm_2O_3$, and $Gd_2O_3$, the catalytic performance could be further enhanced [303]. Especially, GdNi-$Al_2O_3$, which showed comparable activity to LaNi–$Al_2O_3$ and PrNi–$Al_2O_3$, demonstrated the best coke resistance and stability, because it could decrease the graphitization degree of the carbon deposited over the catalysts [303].

### 4.6. Plasma and electrical field enhanced SRM

In steam plasma methane reforming (SPMR), the plasma energy is the excitation source and catalysts are normally not needed [45]. Besides, it allows the production of high-quality gas with controllable composition, shows fast response time and could attain $CH_4$ conversion of 99.5% and $H_2$ selectivity as high as 99%. Furthermore, pollutants emission can be avoided because the plasma decomposes efficiently all complex molecules. Non-thermal plasmas, e.g., cold plasma and warm plasma, which are compact and response fast, are suitable for distributed hydrogen production at small scales [304].

Electro-catalytic reforming was proposed for biomass tar conversion and bio-oil reforming to obtain syngas or hydrogen [46]. The technology has also been extended to SRM to enhanced the reaction efficiency [305]. During electrical-field enhanced SRM, the catalyst bed was equipped with a resistance wire which emitted numerous thermal electrons to realize a synergistic catalysis, thus promoting the catalytic activity and coking resistance. Recently, electro-catalytic SRM over Ni–$CeO_2$/γ-$Al_2O_3$–MgO catalyst exhibited 96.4% $CH_4$ conversion, 75.3% $H_2$ yield, and 40.1% CO selectivity at 600 °C and current of 4.5 A [46]. Thermal electrons could improve the catalytic activity by reducing $Ni^{2+}$ into $Ni^0$ state and enhance the coking resistance by reducing $CO_2$ and $H_2O$ molecules into negative ions such as $CO_2^-$ and $OH^-$, respectively. Another attractive method was applying external electric field to SRM which could suppress the coking formation and lower the reaction temperature [306,307]. Also, low temperature SRM could be enhanced by external electrical field [308–310].

## 5. Summary and outlook

Steam reforming of methane (SRM) is an important industrial process, but the most-widely used commercial SRM catalysts, i.e., Ni catalysts, are suffering from stability issues and strict reaction conditions (high temperature and pressure). Over the years, continuous efforts have been paid to improve the SRM technology including the construction of efficient and cost-effective catalysts and the design of advanced SRM processes.

Hitherto, to enhance the activity and stability of Ni based catalysts, various methods have been applied, including adding promoters, regulating the catalyst structure, developing novel supports, and constructing unsupported Ni catalyst. Ce (Ce-doped Ni/$Al_2O_3$, 75% $CH_4$ conversion), Co (Co-doped Ni/$Al_2O_3$, 95% $CH_4$ conversion), Nb (Nb-doped Ni/$Al_2O_3$, 98% $CH_4$ conversion), K–Ti ($K_2Ti_xO_y$-doped Ni/$Al_2O_3$, 97.2% $CH_4$ conversion), and Ce (Ni/Ce–$ZrO_2$, 97% $CH_4$ conversion) have been proven excellent promoters for Ni based catalysts. NiO–MgO solid solution catalysts ($Ni_{0.4}Mg_{0.6}O$, 100% $CH_4$ conversion; spc-$Ni_{0.5}$/$Mg_{2.5}$Al-aq, 100% $CH_4$ conversion) are especially promising for efficient low-temperature SRM. Despite the reduced loading of noble metal (Ru, Rh, Pd, and Pt) based catalysts with appreciable performance, their high cost limits their development. The other non-noble transition metal catalysts, such as Co–Pt–Zr–La/$Al_2O_3$, Cu/$Co_6Al_2$, $Mo_2C$/$Al_2O_3$, and $Ce_{0.9}Gd_{0.1}O_{2-x}$, have demonstrated great potential, but the sophisticated synthesis could inhibit their future application.

Compared to the catalyst design, developing advanced SRM processes for more efficient and energy-saving conversion of methane to desirable products are more intriguing and effective. As upgrades of the conventional SRM reaction, SESRM (99% $CH_4$ conversion using CaO–$Ca_9Al_6O_{18}$ as the sorbent), OSRM (~100% $CH_4$ conversion over noble metal doped Ni/$Al_2O_3$), CLSRM (99.8% $CH_4$ conversion using Y-doped NiO/SBA-16 as the oxygen carrier), and the newly reported thermo-photo hybrid, plasma and electrical field enhanced SRM technologies, have exhibited their great potential for future utilization. The introduction of sorbent, oxygen, and chemical oxidants can enhance the selectivity of $CH_4$ conversion to CO, reduce the thermal energy input, and increase the reaction efficiency, respectively. The coupling of photo, electrical, and plasma energy with the thermal catalytic procedure not only improved the reaction efficiency, but also could tune the product selectivity. However, challenges remain for the wide application of these methods. For SESRM, it is still challenging to construct a highly stable support that can maximize the interaction between the catalyst and the sorbent. OSRM is limited in its hydrogen production performance. A smart control is required to achieve ideal CLSRM. The reaction efficiency of photo and thermo-photo catalytic SRM needs to be substantially improved. Breakthroughs in catalyst development are required for efficient and robust methane-SOFCs. The development of apparatus for plasma and electrical field enhanced SRM should be paid more attention to enhance the capacity and reduce the overall cost. In addition, following are a few points that could be considered to boost the development of SRM technology for efficient methane utilization.

First, highly efficient and stable catalysts should be explored primarily based on the low-cost Ni and other non-noble metal based catalysts with easily scalable synthesis methods. And their low-temperature performance in conventional SRM and activity in the advanced SRM systems should be especially investigated.

Second, reaction mechanism understanding is essential for reasonable catalyst design. Studies on the mechanism of the diverse SRM processes based on advanced technologies are encouraged. For example, in-situ X-ray diffraction (XRD) and in-situ X-ray photoelectron spectroscopy (XPS) can be applied to investigate the changes of the catalysts and the active species during the reaction process. Also, X-ray absorption near edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) could be applied to reveal the interaction between the composing elements in the catalyst. Furthermore, theoretical calculations should collaborate intimately with experimental work to unveil the phenomena observed during the reaction process.

Also, the advanced SRM processes, especially the newly developed processes with hybrid energy fields (thermo-photo hybrid SRM and electrical field enhanced SRM etc.) have shown great promise, but more attention is needed to drive their practical application. Smart reactor design is especially essential to maximize the synergetic effects of multiple energies. We believe, with the current reliance on hydrogen and syngas production by SRM technology, a prosperous application of the upgraded SRM systems would be witnessed in the near future.

Last but not the least, future efforts should be made on the exploration of more advanced processes with more valuable chemicals (such as methanol) as the major products. In this way, the steps for value-added chemical production can be simplified and the overall energy consumption and capital cost could be dramatically saved. To realize this, novel upgrades of SRM systems can be attempted, for instance, the combination of SRM with other chemical reactions (e.g. steam reforming of bio-oil) and SRM enhanced by external energies (e.g. magnetic force) that can effectively change the selectivity of the catalysts.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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